PHYSMON / RESEARCH PROTOCOL

PHYSICS-COUNTERFACTUAL MECHANISMS

Benchmark Red-Team, DECOY-v2 Reconstruction, and a Gated Mechanistic Research Programme

Revised research redesign, v2.0

14 October 2026

DOCUMENT STATUS

This is a proposed, falsifiable experimental protocol, not a record of experiments that have been performed. Reported Phase-I evidence comes from the actual 25-page ICLR 2027 submission *Confidence Without Cause*. The revised protocol prioritizes benchmark validity and external replication before resource-intensive mechanistic claims.

Maintained as an investigator working document. Revision-sensitive hypotheses, labels, validation decisions, and claim scope require explicit decision-log entries.

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1 Status, scope, and rules of evidence

Purpose. Replace, rather than lightly revise, the June 2026 Phase-II plan with an independently reasoned, falsifiable, executable research programme. **Revision v2.0 adds a mandatory benchmark red-team and reconstruction programme (Phase B), after a critical audit of DECOY-v1. The previous v1.0 research redesign is preserved downstream with revised order and conditionality.** The target is a scientifically strong future ICLR-calibre submission, *not* a guarantee of acceptance. The programme starts from the **actual submitted 25-page ICLR 2027 paper**, *Confidence Without Cause: Predicting and Localizing Physics-Shortcut Sensitivity in Language Model Hidden States* (submission 50608), and treats its experiments as **reported Phase-I findings requiring reproduction**. The prior 30-page proposal, its 14-page supplement, the 42-page Phase-II plan, and the OpenReview/PAT feedback are historical context, not authorities on what must be done next. The differently titled 29-page *Proved, Not Annotated* file has not been established as the submitted manuscript and must not be confused with it.

Critical epistemic constraint. A symbolic certificate that the *correct physical answer* is invariant to cue edits does **not** prove that the model's probability of a particular *answer string* ought to remain invariant; it also does not prove cue use in the model's reasoning. Cue effects on probabilities, internal representations, correct/incorrect answers, and physically faithful computation are separate estimands. A first-rate study must show which estimands move and why. Throughout this protocol: **S** means an observational sensitivity measure; **mechanism** requires controlled, specific, predictive intervention evidence; **shortcut reliance** is reserved for evidence that model computation improperly incorporates the irrelevant cue, not merely a log-probability fluctuation.

Research thesis to test, not presume. Some irrelevant numerical distractors compete with the task-relevant quantity during intermediate processing, and this competition is mediated by a structured and partially generalizable internal pathway. Distinct alternative theses are (i) language/answer-string probability and formatting effects, (ii) generic difficulty or uncertainty, (iii) numerical copying / answer contamination without intervening on the physical solution, (iv) ordinary sensitivity of conditional language-model distributions to context, and (v) multiple unrelated cue-specific mechanisms. All are legitimate outcomes. Do not design experiments whose analysis only accepts the preferred hypothesis.

Scope strategy. An upstream **Phase B (B0–B7)** tests whether the benchmark measures a compelling, valid phenomenon and decides whether to rebuild, narrow, or pivot. The later mechanistic **critical path** contains twelve blocks (P0–P11), some of which implement or deepen Phase-B checks rather than starting a separate dataset enterprise. A subsequent research catalogue gives compulsory controls, secondary branches, and high-upside extensions. Some experiments are mutually exclusive alternatives or conditional escalations; performing every possible experiment is neither mathematically achievable nor scientifically optimal. Maximize discriminating evidence per unit of model compute and human review, while retaining a documented broad universe of possibilities. No new unvalidated experiments should be smuggled into a headline claim after seeing outcomes.

Baseline source categories. `SUBMITTED` = reported in the actual ICLR manuscript, `REVIEW` = criticism appearing in the supplied PAT review, `LITERATURE` = independently checked public work linked in References, and `PROPOSAL` = a new suggested experiment, hypothesis, threshold, or resource allocation. Numerical targets in this document other than clearly marked SUBMITTED values are **planning choices, not results**. Repository outputs are not attached here; therefore this plan cannot verify actual raw data, scheduler jobs, code, or version history.

2 Executive scientific judgment

The Phase-I work has potentially valuable ingredients, but the next paper needs *one of two qualitatively stronger outcomes*:

Track M (mechanism-first, conditional on passing benchmark validity gates**, preferred only if supported):** a verified causal account of a measurable cue-related computation. It should explain *what*

information from the distractor is represented, *where it travels*, *how it changes the competition between a task-relevant answer and a distractor-derived alternative*, and *which intervention* selectively removes that dependence while preserving correct physics. Its predictions must succeed on unseen physics templates and intervention sites, with a later replication in an independently selected model. A single exceptional head, even with a large recovery value, is insufficient.

Track E (evaluation/measurement, fallback with genuine scientific value): a rigorous decomposition showing which previously apparent shortcut signals are artifacts of reference-answer scoring, prompt formatting, metric asymmetry, or data structure, and which survive as reproducible behavioral invariance violations. This could contribute a stronger certificate-backed evaluation protocol and a validated account of the limits of pre-generation monitors. It should not claim a reasoning circuit.

Optional Track R (robustness/repair): a selective, mechanism-informed intervention with a compelling trade-off: improved cue invariance and/or accuracy on unseen data, little damage to legitimate variable use, and an advantage over cheap prompt editing, multi-query verification, and training-based augmentation. This track becomes a paper-defining claim only if it survives held-out evaluation.

Benchmark decision precedes the following decisions: Retain DECOY-v1 for history; rebuild a separate verified DECOY-v2 for controlled physical relevance, remove Cue C from the *irrelevant-cue* core into a representation-invariance track, and require a genuinely external validation set. **Three decisions that precede expensive mechanism work:** (1) Does the claimed behavior survive scoring and task-formulation challenges? (2) Is any internal effect specific to physically irrelevant distractors rather than a generic output perturbation? (3) Does the identified computation generalize beyond the selection panel? If any answer is no, stop using the corresponding stronger vocabulary.

2.1 Verified Phase-I starting ledger (SUBMITTED, not replicated here)

Evidence item	Submitted result / caveat	Consequence
DECOY size	155 canonical four-rendering families; 140 original behavioral families; 15 later-built transfer families	Distinct populations; report separate denominators and version timestamps.
Monitoring panel	135 families: 70 sensitive, 65 stable; five stable original-panel families removed after parsing failures	Primary S_lp nominally does not need free-generation parsing; audit selection bias.
Models	Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, DeepSeek-R1-Distill-Qwen-32B	The reported DeepSeek teacher-forcing position has an unresolved reasoning-trace caveat.
Behavioral sensitivity	Qwen 70/140 at 0.5 nat, Llama 32/140; 25 shared positives	Model dependence is observed under one measurement definition.
Mean-state prompt monitor	L18 AUROC 0.7596 using **four rendering activations averaged**	Four input passes are needed to score one family as defined.
Single-rendering monitor	L16 AUROC 0.7465, separately selected	This, rather than 0.760, supports single-prompt deployment claims.
Strong inexpensive comparator	Fitted next-token entropy AUROC 0.7270; Qwen correct-answer correctness probe AUROC 0.7657	Incremental hidden-state value unproven against best fitted comparators.
Paired statistical comparison	Monitor-minus-entropy +0.033; 95% family bootstrap [-0.074,+0.141]	Does not reject no benefit. The combined-plus-hidden gain +0.067 also spans zero.
Calibration and operating point	Monitor Brier 0.2036 vs entropy 0.2140; entropy ECE 0.0907 vs monitor 0.1041; FNR@FPR=.10 0.643 entropy vs 0.714 monitor	No unqualified ‘better calibrated’ or ‘better operationally’ claim.

Evidence item	Submitted result / caveat	Consequence
Frozen expansion	Pooled AUROC 0.820 on 15 families; within B ~0.667 (n=5) and C ~0.6875 (n=10)	Pooled gain is compatible with cue-type/base-rate confounding.
Qwen causal intervention	L16 four-head knockout recovery 1.870 on 20 selected Cue-B families; head 11 alone 1.848; clipped four-head recovery 0.969	Dominant site, but over-reversal and denominator effects demand absolute metrics and specificity.
Causal transfer	Cue A much stronger than Cue C; DeepSeek tested site small; Llama exploratory later head-2 result not paper-eligible	Do not assume shared circuit or cross-model null.
Validation	Dual independent expert human validation **not completed**	Physics-natural-language contract not independently established.
Structural overlap	34 template clusters, 147 family pairs unresolved, 2,352 candidate rendering pairs; 71% family-held-out tests share a cluster with training	Family LOOCV is not equivalent to template-cluster generalization.
Provenance	182 science runs, registry coverage 0.703 and strict coverage 0.451	Establish a claim-to-artifact ledger before extending.
Renamed-donor control	Zero evaluable rows due to donor-selection defect	**Unmeasured**, not a favorable negative control.

Audit alert A. The submitted paper reports that every Qwen and Llama output in the highlighted CM_B_UM_059 example is wrong, while confidence in the teacher-forced correct answer varies strongly. That is a powerful distinction: the intervention may change the likelihood of an answer string without changing whether the model solves the problem. The research must not call this 'correctness repair' without demonstrating changed free-generation accuracy.

Audit alert B. The manuscript identifies a numerical near-match tail but also reports Pearson r=0.074 over 100 Cue-B families. Near-match *events* are an interesting hypothesis; a general monotone proximity law is not established.

Audit alert C. A nearby older proposal and another manuscript use different naming and evidence-status language. Build a single timestamped authoritative evidence table; never transfer an unverified 'completed' label between documents.

3 A mandatory new Phase B: red-team, reconstruct, and independently validate DECOY

Board decision (14 October 2026). Do not use the existing 155-family DECOY benchmark unchanged as the sole scientific foundation of a future PhysMon mechanism claim. Do not discard its Phase-I evidence, and do not substitute a pre-existing public physics benchmark wholesale. Maintain immutable **DECOY-v1** (historical reproduction and failure diagnosis); conditionally build a **DECOY-v2 experimental suite** for controlled causal discovery; establish a separately sourced, separately certified **External Physics Counterfactuals (EPC)** collection for confirmation; and use **GSM-DC / GSM-Symbolic** as non-physics controls. Treat unit/re-expression invariance as a distinct task, not interchangeable with irrelevant numerical cues. Neither a large newly generated dataset nor improved AUROC makes the work valuable unless these tests establish a precise, nontrivial *model behavior* and an independently validated *causal explanation*.

Scientific reasons for the decision. (i) The original signed base-relative sensitivity label can change when the exact same four renderings are reindexed. (ii) A certificate about a symbolic *answer value* does not constrain the probability of a chosen *answer string*, and physical assumptions asserted in

natural language are not independently checked by free-symbol elimination. (iii) Cue C changes how a *governing* quantity is expressed; it is not the same as adding physically irrelevant information. (iv) Five original Cue C families, 34 structure clusters, 2,352 unresolved cross-family rendering pairs, no completed independent expert validation, and a 135-family monitoring cohort with outcome-related filtering limit inference. (v) Existing GSM-Symbolic and GSM-DC already establish controlled irrelevant-context tests; novelty cannot be 'symbolic templates plus distractors'. Each is a **reported property or a falsification target**, not a finding that the proposed v2 experiments have succeeded.

Benchmark architecture, defined before collecting new outcomes:

Track	Scientific target	Contents / status	Allowed inference
V1-R: frozen DECOY-v1	Reproduction and failure analysis	All 155 original families, 140 Phase-I behavioral families, five monitoring exclusions, 15 later-created transfer families; original signed labels preserved	Exactly which historic claims reproduce under their original protocol; **not** prospective generalization to independent tasks.
V2-A: causal relevance and numeric interference	Whether model use of a numeral is appropriately conditioned on physical relevance	New independently defined physics structures, matched relevant and irrelevant edits, no-cue anchors, nonnumeric/numeric controls, numeral/units factorial	Controlled relevance-selectivity and cue-interference effects on *specified* outcomes.
V2-B: invariance/equivariance	Whether models respect transformations of relevant representations	Equivalent units/coordinate systems, reference-frame transformations where admissible, dimension/precision stress tests; retain old Cue C here	Transformation-consistency failure; **not** by itself irrelevant-cue shortcut reliance.
EPC: external physics counterfactuals	External realism, independent source and structure generalization	Only validated adaptations of PHYSICS-Verified, UGPhysics, selected PhysReason/SciBench cases and investigator-independent expert-authored cases	Generalization beyond v2 template grammar, subject to source-sampling and adaptation constraints.
MATH-X: non-physics controls	Separate physics-specific relevance from generic numerical distraction	GSM-DC dependency graphs, GSM-Symbolic NoOp, unit-stripped or isomorphic algebraic analogues	Whether effects or mechanisms are domain-general.

Strict version policy. Any semantic revision of a governing equation, assumptions, target variable, accepted answer, or cue role increments the family's version; it cannot be silently edited in place. Freeze source dataset revisions and hashes. Once a confirmatory panel is opened, do not use it to tune benchmark generation rules, architecture, scoring, parsers, threshold, site selection, or eligibility. Both negative and ineligible examples remain in an immutable manifest with reasons.

3.1 B0. A benchmark threat model, estimand map, and external novelty audit

B0.1 Define what the benchmark is *not*. The ground-truth theorem is about the physical mapping $y^* = f(r, a)$; it is not a theorem about equal conditional language-model probabilities $p_\theta(\text{string}(y^*) \mid x_i, q)$. An answer-invariant pair establishes a correct *response target*, not the model's actual computation or its internal dependence on a nuisance. No label should automatically be named `shortcut_reliance` without controlled behavior and mechanism evidence. Declare separate records for (a) physical target invariant, (b) observed gold-string drift, (c) actual-answer shift, (d) distractor-derived answer preference, (e) erroneous dependency on a physical cue, (f) causal internal component or pathway.

B0.2 Formal object. A task specification stores physical state s, assumptions a, a dependency graph/program g, target t, physical answer $y^*(s, a, t)$, candidate cue z, semantic role $R(z)$, perturbation operator I, and rendering $\mathcal{R}$. A nuisance intervention satisfies $y^*(I_z(s), a, t) = y^*(s, a, t)$ within a declared domain $\mathcal{D}$; a relevant intervention is a *positive control* for which the expected transformed target $y^*(I_r(s), a, t)$ is computed, including cases where the numerical answer happens to coincide despite true causal dependence. The role cannot be established by answer-value coincidence alone: a variable can be causally relevant even at a parameter setting with zero local effect.

B0.3 Preregister the threat taxonomy. Audits must attack: incorrect physical semantics, missing boundary conditions, excluded-domain exceptions, dimensional/unit errors, answer-rounding ambiguities, canonical-string/tokens, no-cue anchor dependence, arbitrary base choice, irrelevant numerical *value* versus extra text, model prompt wrapper and generation position, template or source overlap, near-match selection, cue-condition class prevalence, sampling/decoder variation, annotator suggestion/automation bias, architecture-specific scoring incompatibility, adaptive head selection, data contamination, and publication novelty. Each risk receives a test, accountable owner, possible failure class, and action.

B0.4 Closest-work matrix. Inspect **GSM-Symbolic** (ICLR 2025), **GSM-DC** (EMNLP 2025), **UGPhysics** (ICML 2025), **PhysReason** (ACL 2025), **PHYSICS-Verified** (audited public release), and relevant mechanism papers. For each: dependency program, control of nuisance/relevant edits, formal guarantee, human semantics review, challenge difficulty, model-output metrics, intervention capability, public data/derivation, and external transfer. The proposed differentiator must be independently confirmed as a *joint contribution*: physical-role interventions + honest behavior endpoints + mechanistic prediction/causal rescue + external confirmation. Do not call v2 novel because it is larger than v1.

Deliverable B0: a versioned `benchmark_claims_and_threats.md`, a closest-work table with accessible paper/repository URLs, an estimand glossary, and an independent red-team protocol. **Failure branch:** if physics relevance cannot be distinguished from GSM-DC-style dependency selection, frame the research as generic numerical interference with physics as a controlled domain rather than claiming unique physics reasoning circuitry.

3.2 B1. Reproduce and deliberately break the original benchmark

B1.1 Complete provenance reconstruction. For **every** v1 rendering, retrieve exact solver expression/assumption code; units/target; cue value; base index; prompt bytes; answer string; token IDs; native chat template; generation settings; raw log-probability; raw output; parse status; original label; model commit; split/cluster; and intervention metadata. Reproduce every 155/140/135 denominator and original Phase-I table without filling missing values. Identify which of the 15 new families belong to later confirmation, not unlabeled failed examples. Record author/agent creation, numeric-instantiation parentage, and original inclusion/exclusion timestamps.

B1.2 Permutation attack (no GPU needed if raw values exist). For each four-rendering family, compute the legacy signed sensitivity under **all four possible choices of base**; its minimum, maximum, number of positive labels, fraction whose label flips, and pairwise rank inversions across models. Report overall and separately for Cue A/B/C, law, answer-token length, original sensitivity quantiles and family cluster. Compare with symmetric range $\max \ell - \min \ell$, median/SD/MAD of ℓ_i,

and truly anchored no-cue contrasts (which *do* require new evaluations). Preserve original threshold-registration history; do not describe new metrics as having been preregistered in June.

B1.3 Submitted-example sanity test. The reported torque family has Qwen gold-string log probabilities $(-45.531, -55.562, -45.000, -46.469)$; base 0 yields about $+10.031$ nats whereas base 1 yields about -9.093 nats. Write an executable unit test for this exact result. This is a **mathematical vulnerability of the chosen statistic**, not evidence that the underlying model effect disappears with reindexing. Do not falsely claim that the order-independent range is a signed risk measure: it is absolute dispersion and must be interpreted separately.

B1.4 Source audit of generated and ambiguous answers. Check whether the primary teacher-forced measure actually requires free-generation parsing. Inspect why the five stable families were excluded from the 135-family monitor, recompute inclusion-agnostic analyses, and treat parser failures as their own observable. Validate example claims of correct/free-generated responses against raw outputs; enumerate the `wrong->same wrong`, `wrong->different wrong`, and `correct->wrong` cases without cherry-picking. Test the effect of source answer variants, precision and units on likelihoods; run targeted manual scoring on original large-effect families.

B1.5 Certification stress audit. Review all symbol checks and unit identities under changed assumptions: implicit friction, nonzero mass of a pulley, coupling through shared field, feedback/interaction with an 'isolated' system, sign and direction conventions, frames and unit conversion, piecewise conditions, singularities, domain restrictions, physically nonpositive parameters. A certificate that a nuisance symbol is absent from the *chosen* equation is sound only given the equation and assumptions; test whether the natural language provides them. Use positive mutation controls that must remain valid and negative mutations that must fail.

B1.6 Structural overlap and authorship. Reconstruct the reported 34 clusters/147 family pairs/2,352 unresolved rendering pairs, audit what 'unresolved' means, review overlap graph edges rather than declaring all collisions. Re-evaluate family-held-out results using unioned symbolic-computation-cluster splits and, separately, law-class and problem-source splits. Test whether the strongest non-activation predictors can detect the purported label from template ID, governing-law metadata or crude quantity features.

Gate B1: publish an audit dashboard of verified, invalid, semantic-ambiguous, parser-uncertain and missing records, plus a table of metric/base/wrapper robustness. If the Phase-I effect remains only a teacher-forced string effect, preserve it as such; **no new claim that model physics reasoning was manipulated**. If benchmark artifacts cannot be reproduced, stop all dependent causal claims pending provenance repair.

3.3 B2. Decide the correct construct before choosing how many examples to generate

B2.1 Define four different hypothesis families. (H-P) Physical relevance selection: does the model correctly exclude non-governing physical variables and include governing variables? **(H-N)** Numerical interference: does magnitude/precision/digit identity of an irrelevant cue compete with target magnitude selection? **(H-T)** Text/interface effects: does number placement or tokenization alter a single reference-answer string without changing the physics solution? **(H-U)** Unit/representation transformations: does the model represent equivalent governing quantities consistently? Do not pool the fourth with the first three.

B2.2 Specify primary behavior as an interpretable contrast. For each physically matched base task, observe deterministic answer correctness and answer class under (i) no cue, (ii) irrelevant cue of varying magnitude, (iii) style-matched nonnumerical extra content, (iv) generic numerical metadata, (v) physically relevant edit with recomputed gold answer, and (vi) adversarially constructed decoy answer. Primary recommended co-endpoints for the *pilot*: difference in probability of a wrong **decoy-derived** candidate under a fixed, prespecified candidate space, and change in free-answer correctness / invalid answer rate; report symmetric teacher-forced dispersion as a diagnostic. Select one primary endpoint for confirmatory testing only after pilot reliability and incidence have been estimated.

B2.3 Relevance is not a keyword. Match tasks with *explicitly disconnected* and *physically connected* subsystems, but do not fabricate a 'single-word role flip' if it changes other physical assumptions or allows multiple plausible interpretations. Every pair gets its own formal state change, recomputed solution and expert semantic validation. Add *subtle* relevance requiring causal tracing (an intermediate variable changes only through an indirect coupling) alongside explicitly stated irrelevant cases, to distinguish genuine relevance computation from memorizing `isolated` or `ignore`.

B2.4 Establish four eligibility strata before scoring. All valid problems (intent-to-evaluate population); **format-parseable** (limited to metrics truly requiring parsing); **base-correct** (what happens to a model already solving a task); and **model-positive** (eligible for sensitivity reduction or causal recovery). Report each, with its denominator and selection mechanism. A result conditional on base correctness is not a general result about all questions. The original signed effect must never be substituted for the question 'does a model output an incorrect answer?'.

B2.5 Fix metric calibration and negative controls. Include symmetric range, actual answer flip and correctness, right-vs-decoy log odds, first-token/full-sequence scores and varied semantically equivalent output strings. Define a permitted null distribution of cue-free and formatting-only changes. Report *absolute* differences and robust quantiles; do not rank architectures solely by proportion crossing 0.5 nats. A no-cue reference is scientifically meaningful but cannot be called a universal neutral condition without testing omission-versus-added-text asymmetry.

Gate B2: If H-P or H-N produces reliable behavior beyond generic text and answer-string effects, proceed to a narrow causal benchmark emphasizing that phenomenon. If only H-T survives, create a measurement/LM-interface project and freeze physics-reliance claims. If only H-U survives, split it into a formal *equivariance* study, not the core shortcut programme. If all effects disappear under realistic controls, archive as a carefully documented negative result and do not manufacture additional distractor conditions to force a positive headline.

3.4 B3. A modular v2 benchmark with mutually discriminating controls

B3.1 Five problem-structure strata. (i) Single governing equation with natural nuisance (calibration/causal simplicity); (ii) multistep dependency graph with the distractor attached to a non-ancestor node; (iii) interacting subsystems with an explicit or inferable physical disconnection; (iv) conditional laws/boundary constraints where relevance depends on a stated regime; (v) counterfactual 'role-reversal' examples where the same measured quantity is relevant in one verified world and irrelevant in another. Include inverse problems and sign/vector questions where answer and representation validation are reliable. **Do not assume** thermodynamics or multimodal diagrams are easy to certify: admit only validated examples and retain excluded candidates in a selection log.

B3.2 Typed executable graph generator. Each node stores dimension, unit, reference frame, sign convention, physical quantity type, provenance (measured/computed/assumed), uncertainty where relevant, and parent edges. Include a solver program and a **small explanation-free proof object**: assumptions, symbolic identities, relevance/non-relevance certificate, domain of admissible cue values and tested boundary cases. Maintain multiple routes to verify selected hard instances (e.g., independent numerical solver or human algebra), not merely multiple outputs of the same symbolic equation generator.

B3.3 Core 3-factor behavioral interaction. Cross **physical relevance** (governing vs non-governing), **numerical proximity** (near-match vs predeclared far band) and **unit relationship** (same/equivalent vs incompatible); other factors held balanced. Primary statistical target is a preregistered interaction in *answer competition or correctness*, with cluster-level inference. Add diagnostic cells that isolate **digit overlap** from mathematical magnitude: close magnitudes with little lexical overlap; far magnitudes with shared leading digits; numerical equality but incompatible units; unit-compatible far distractor; unit-incompatible but lexically overlapping distractor. Treat not all unit/relevance combinations as physically admissible; use design constraints and explicitly record empty or impossible cells rather than inserting nonsensical prompts.

B3.4 Factor families to expand only after the core signal. Placement/recency, ordering, number of nuisance values, repeated mentions, prose versus table/diagram, explicitness of isolation, negation, irrelevant explanation length, tokenizer count, decimal precision, rounding, numerical sign and extremes, time-scale, unit scaling, reference-frame change, numerical instantiation, paraphrase and language, law familiarity, chain length, distractor attached to intermediate versus final target, multiple candidate governing laws, and contradiction detection. For each secondary factor define a *paired negative control* and at least one alternative causal explanation. Don't claim a full Cartesian product: use balanced incomplete blocks or fractional factorials with estimable interactions and a documented design matrix.

B3.5 Control no-op robustness vs valid sensitivity. Each family must have at least one *irrelevant* edit whose target is invariant and, wherever feasible, a matched *relevant* edit with a known changed solution. Also test a coupled-system counterexample where the formerly irrelevant physical quantity becomes governing; a model that simply ignores **all** extra numbers cannot pass both tests. Score the relevant positive controls on correctness of the *new* target, not 'output invariance'. Include deliberately ill-posed/under-specified prompts solely in a labeled rejection task, never as verified-invariant examples.

B3.6 Distributions matter. Maintain separately (a) unbiased/naturalistic distractor sampling, (b) balanced experimental intervention cells, and (c) budget-constrained adversarial optimization. Freeze the first two before seeing model scores; adaptive search examples must carry attack budget, query history, target model and generation seed. Never estimate typical failure rates from an adversarially optimized subset. Hold out newly generated source graphs and cue-value random seeds for final test.

B3.7 Difficulty and ecological validity. Benchmark difficulty with *independently measured* physical structure: minimum derivation steps, unique dependency nodes, number of applicable laws, number of constraints and domain conditions. Supplement with blinded human solve-time/difficulty assessment. A model's low accuracy does not establish higher structural difficulty. Include problems with ordinary narrative plausibility; separate unnatural constructed minimum pairs for mechanism discovery from human-readable external source questions for final evidence.

Deliverable B3: proof-carrying generator; a fully specified factorial design matrix with feasibility flags; source/variant lineage; formal invariance+positive-control tests; first 10 wholly new certified structures and a **discovery-only** sample of roughly 40–60 independent structures for the gate pilot. **Failure:** artificiality or impossibility of role-matched cells -> redesign the generative grammar or narrow the causal question, not loosen verification criteria.

3.5 B4. Human validation as an independent scientific experiment

B4.1 Audit v1 before claiming certification of English text. All 155 families receive at least one qualified blinded semantics review; all families used in a publishable confirmatory claim receive **two independent** reviews and third-party adjudication where needed. The validation pilot includes samples from A/B/C, all three original physics domains, largest gold-likelihood shifts, negative scores, presumed easy stable cases, ambiguous coupled systems, and high-overlap clusters. Do not let the researchers who wrote a family silently self-certify its semantic naturalness.

B4.2 A two-pass review instrument. Pass 1, without the authors' equation: solve or outline the applicable physics; state assumptions, target, whether the cue could affect target, and whether wording permits another physical reading. **Pass 2, with the encoded equation/certificate:** check consistency of graph, units, domain, numerical answer, specified cues and any omitted dependency. Separate validity, clarity, difficulty and naturalness judgments rather than collapsing into one yes/no. Require validators to mark uncertainty explicitly and annotate counterexamples.

B4.3 Qualification and independence. Screen reviewers with calibrated physics problems including boundary-condition traps, unit conversions and deliberately misleading 'isolated' claims; record expertise by domain. Balance assignments across authors, source datasets, cue types and expected difficulty; reviewers are blind to model outcomes and probe/circuit hypotheses. Estimate payment/time during a small pilot, then resource the whole set. Disagreements are adjudicated

by a domain-qualified independent third reviewer. Agreement metrics include raw agreement, Cohen's kappa on substantive claims, uncertainty intervals, and category-specific defect rates; report unresolved ambiguities.

B4.4 Adversarial validation tests. Before release, insert **predeclared planted defects**: hidden coupling, missing assumption, relevant variable removed from equation, failed transformation identity, inconvertible or inconsistent units, invalid parameter domain, insufficient boundary condition, invalid numerical precision, contradictory distractor statement, and a valid equivalent rewrite. Validate detection sensitivity and false-rejection rate by mutation type. If reviewers systematically miss a defect type, revise the checklist and repeat calibration before confirmation proceeds. The planted-invalid bank never enters the benchmark as 'genuine' physics.

B4.5 Admission decision categories. `CERTIFIED_AND_VALIDATED`, `SOLVER_VALID_LANGUAGE_AMBIGUOUS`, `SOLVER_INVALID`, `ANSWER_AMBIGUOUS`, `NATURALNESS_FLAG`, `REVISED_REVALIDATED`, `EXCLUDED`, and `UNREVIEWED`. Report fractions both before and after repair. Store a signed reviewer record; do not silently count repaired examples as untouched independent examples. Publish a human-audit protocol and aggregate outcomes without exposing reviewers' personal identifiers.

Gate B4: Confirmatory physical-relevance claims are blocked until independent expert semantic validation, version-correct solver checks and the parser criterion for the relevant endpoint pass. No arbitrary agreement threshold substitutes for reviewing concrete physics defects. High disagreement or mutation-test failure means a new validation cycle, not optimistic averaging.

3.6 B5. External datasets as source material, not borrowed certificates

Why the hybrid design matters. Public benchmarks supply diversity and independent original problem authorship, but generally **do not supply DECOY-specific nuisance/relevance interventions or proofs of our modified problem's target**. The moment a source problem is rewritten, the adaptation becomes *new benchmark data requiring independent verification*. Synthetic v2 and adapted public questions serve different causal and generalization purposes.

Candidate source	Publicly established properties	Proposed adaptation / main danger	Decision
PHYSICS-Verified (Yale NLP)	Public **1,034 PhD-qualifying-exam problems / 2,559 individually scored answers**, cleaned reference solutions; six physics areas; source card recommends it for new evaluations	High structural/difficulty realism. Source answer checks **do not validate our newly inserted cue**, and many proof/multimodal tasks are excluded. Small models may be mostly incorrect on untreated advanced items.	**Primary hard external source**, screened for tractable symbolic/numeric intervention candidates.
UGPhysics (ICML 2025)	**5,520 undergraduate questions**, 13 subjects, English/Chinese, seven answer types	Strong breadth; varying answer-type and multimodal complexity; source license and derived-work permissions need review; not every question supports a certifiable intervention.	**Primary broad external source** with predefined per-subject selection record.
PhysReason (ACL 2025)	1,200 physics problems; knowledge/reasoning strata and process evaluation	Useful multistep dependency-path challenges; reference rationale is not an internal reasoning trace and may not imply verified nuisance role.	**Secondary deep-structure source** after source-problem reconstruction.

Candidate source	Publicly established properties	Proposed adaptation / main danger	Decision
SciBench (ICML 2024)	College-level scientific tasks from course material	May contain unsuitable questions/figures or source-specific copyright constraints; new interventions require review.	Optional external source when adaptation feasibility is demonstrated.
GSM-DC (EMNLP 2025)	Controlled symbolic dependency graphs, distracting-context injection and stepwise accuracy analysis	Highly similar behavioral confounds; mathematics rather than physics	**Required methodological competitor and cross-domain control**, not replacement.
GSM-Symbolic (ICLR 2025)	Symbolic variants and irrelevant NoOp clauses in math	Demonstrates numerical/context fragility; does not automatically establish a PhysMon circuit	**Required pre-existing benchmark comparator.**

B5.1 Sample the candidate pool independently of model outcomes. For each source, document dataset revision, inclusion criteria, population, random seed, subject/answer-type strata, sampling probabilities, original ID and provenance, any figures removed, and open-access/reuse constraints. Do *not* first inspect which published problems Qwen fails and preferentially adapt the dramatic ones for a 'representative' external test.

B5.2 Conversion pipeline. (1) Identify the original target answer and material assumptions; (2) solve independently or reconstruct a transparent executable dependency program; (3) identify safe intervention points and candidate irrelevant/relevant quantities; (4) derive the output under both edits; (5) preserve essential problem semantics when rewriting; (6) conduct independent physics and naturalness review on *each adapted pair*; (7) audit source overlap with v1/v2 and model-training contamination risks; (8) freeze result-bearing adapted examples and scoring before evaluating models. Log reason-coded rejections at *each* step.

B5.3 Feasibility pilot. Draw a **precommitted, stratified 20–40-candidate sample from each of PHYSICS-Verified and UGPhysics**, with an optional smaller PhysReason sample; measure eligible-adaptation yield, reviewer hours, formal-certification rate, question difficulty and baseline solve rate. Report uncertainty intervals on yield and compare representativeness of retained/rejected questions. This is not yet a final external test; it determines whether each source is appropriate.

B5.4 Two independent external sets. (i) **EPC-D**, an openly available adapted-development set that can inform *methods and tooling* but never confirm an already tuned claim; (ii) **EPC-C**, a locked external confirmation set authored/adapted/verified by different staff after the mechanism's target and code are frozen. Aim for heterogeneity across law, source and difficulty, not merely a larger number of paraphrases. Separately maintain **EPC-N**, a new set of expert-authored problems not derived from public benchmarks to help limit contamination and copied-reference leakage. The final external set is opened once, and any post-failure revisions require a new independent set.

B5.5 Source- and solvability-aware estimands. Report all eligible cases, base-correct cases and available physically meaningful wrong-answer alternatives. Small models will plausibly fail more difficult problems; in that event, don't cherry-pick only successes without disclosing selection and shifted target population. Use relevant-edit responsiveness even for models without baseline correctness, and stratified analyses where scientifically interpretable. Verify the native unit/answer scoring for each model before comparing them.

B5.6 Cross-domain falsification. Select abstractly matched GSM-DC and GSM-Symbolic structures, measure interference under their own validated protocols, then evaluate a carefully translated subset

with PhysMon's controlled relevance conditions. If a supposed 'physics relevance gate' responds identically to graph-equivalent grade-school arithmetic, interpret the mechanism as general dependency filtering or numerical copying unless a truly physics-specific manipulation distinguishes it.

Gate B5: If adaptations are rare/ambiguous, preserve source rejection rates and either restrict the external question types or commission independent expert-authored tasks; don't downgrade certification to inflate numbers. If only synthetic tasks show effects, make that limitation central and do not claim broad physics reasoning. Confirmed external effects are required for a broad ICLR-level generalization claim.

3.7 B6. Pilot, power, confirmation, and anti-cherry-picking rules

B6.1 Pilot matrix and endpoints. Run roughly **40–60 new independent** discovery structures, enough to debug the construction and estimate variance, not to prove a small effect. For each, balance a prespecified attainable subset of relevance × proximity × units, no-cue and numeric metadata conditions; include at least two numeric instantiations and a fixed output grammar. Two independently implemented dense models are a defensible starting point; a third stronger model tests whether the phenomenon persists beyond 7–8B without presuming it will. Preselect the primary cue family and endpoint only on this discovery partition.

B6.2 Estimate precision at the correct independent unit. Perform simulations and blinded design checks with random effects for physical structure, source/author, numeric instantiation and model. Target CIs for paired accuracy differences, relevance×proximity interactions, out-of-cluster AUROC margins, and causal rescue effect. Use a power target and minimum effect size that have an engineering/scientific interpretation. The pre-existing '300–600 symbolic structures' is a **tentative upper planning band**, not a quota or evidence that 600 will have enough independent laws. A large library of paraphrases can be statistically less informative than a smaller set of independent dependency programs.

B6.3 Lock five disjoint roles. `EXPLORATION` (pilot and unrestricted scans); `GENERATOR_TUNING` (naturalness and control debugging, still nonconfirmatory); `INTERNAL_CONFIRMATION` (never used for choosing labels, models, heads or thresholds); `EPC_CONFIRMATION` (independent external source); and `ADVERSARIAL_CHALLENGE` (separate worst-case claim). Merge/group by graph lineage and template structure before splitting. When physically related groups cannot be separated without losing power, state the limitation and reduce the generalization claim.

B6.4 Primary hypothesis and the factorial null. Example prospective primary: a non-governing same-unit near-match alters **correct-versus-decoy log-odds** more than a token- and length-matched far distractor, while a physically *governing* change is tracked correctly. The exact statistical interaction and candidate-answer set must be frozen with the model and prompting regime. Declare an alternative hypothesis for lexical copying and a predeclared 'no practical effect' interval. Family-cluster paired tests, hierarchical regression and within-cluster permutation tests should agree qualitatively before calling the finding stable.

B6.5 Reliability and adversarial budget. Reuse fixed cue random seeds across models; repeat deterministic scoring, investigate numerical precision and GPU backend variation, sample free generation where appropriate, and report median/tail outcomes. For adaptive adversarial cue search, fix allowed edit space and per-family query budget; report attack success **separately** from fixed-distribution robustness. Avoid training monitor or selecting causal heads on adversarial-targeted test examples.

Gate B6: Promote to heavy mechanistic research only when a validated causal contrast exists on held-out structural units, is not dominated by format/tokenization or one cue subtype, and is large enough to support meaningful interventions. If effect is absent on two independent sources, stop the broad claim and report boundary conditions. If model output shifts only within teacher-forced string probabilities, restrict the target to a calibrated likelihood/interface mechanism rather than physical shortcut reliance.

3.8 B7. Benchmark deliverables and acceptance rules for the later mechanism study

Release package. `decoy_v1_frozen_manifest` (legacy IDs and exact original scoring); `decoy_v2_structures` (typed graphs, physical assumptions, symbolic/numeric solvers, invariant and relevant-edit certificates); `decoy_v2_renderings` (all factor cells, random generators, source lineage and text hashes); `validation` (blinded decisions and adjudication); `scoring` (model/wrapper/answer-token IDs and parser versions); `external_adaptations` (source IDs, acceptance/rejection ledger, licenses and validator decisions); `splits` (lineage/cluster firewall); `controls` (no-cue, neutral text, generic numerals, role reversals, ill-posed negatives); `results` (exact estimands, eligible denominators, uncertainty and all failed runs); `benchmark_card` (intended uses, disallowed inferences, known limits, data provenance, contamination). Exact path strings are illustrative; finalize naming after code inspection.

Quality-gate checklist: independent correct physics; explicit assumption and domain; proof or independent executable certificate; verified relevant and irrelevant roles; matched surface manipulations as specified; validated parser and answer equivalence; source/license traceability; independent blinded review; cluster-independent split; frozen intervention budget and scoring; balanced reported strata; all intended-to-evaluate records; unchanged phase-1 reproduction; and one untouched external sample. No 'certified' claim about natural-language semantics based solely on a symbolic free-symbol test.

Benchmark-only scientific value test. If the rebuilt dataset does not enable a genuinely new comparison that GSM-DC, GSM-Symbolic and existing physics benchmarks cannot support, **do not pitch benchmark size or formal verification as the lead novelty**. Use it as infrastructure for the causal mechanism paper. A standalone dataset contribution requires demonstrated measurement validity, distinctive physics relevance factors, independently validated external tasks, baseline coverage and usable release tooling. Compare cost and quality, not only sample counts.

Final Phase-B go/no-go: `GO–MECHANISM` if physically validated, structure-generalizable, answer-relevant interference is reproducible and sufficiently powered; `GO–MEASUREMENT` if a coherent output-likelihood/representation-invariance phenomenon survives but correctness/relevance claims do not; `PIVOT–GENERAL NUMERIC` if analogous GSM-DC/physics-stripped tasks explain the effect; `REDESIGN–BENCHMARK` if natural-language semantics or scoring cannot be trusted; `STOP BROAD CLAIM` if no meaningful effect survives held-out validation. Gate outcomes do not license retroactive relabeling of confirmatory data.

4 The falsifiable theoretical framework

Let a physical task be `T=(r,a,g,y*)`, with relevant variables `r`, assumptions `a`, governing equation `g`, certified answer `y*`, and nuisance cue `z`. A well-designed family contains independently varied cue `z`, numerical relevant variables, wording, output wrapper, and cue-placement/representation factors while recording what is held fixed. `Y*(x)` is physically invariant for certified nuisance edits. The model has a distribution `p_theta(.|x,q)` depending on prompt `x` and answer-scoring protocol `q`; never omit `q` when studying probabilities.

Observable endpoints (distinct by construction):

1. `C_gen`: free-generation correctness under an independently audited parser, with abstentions and invalid outputs explicit.
2. `A_gen`: normalized parsed free-generation answer; `F_flip`: pairwise answer disagreement within family, stratified by correctness transitions.
3. `L_gold(x,q)`: length-normalized and raw teacher-forced log-probability assigned to a *specified canonical answer sequence* under protocol `q`.
4. `D_lp`: distribution of within-family gold-answer likelihoods; primary reference-free candidate `max_i L_gold - min_i L_gold`, accompanied by SD/MAD and pairwise absolute differences. Unlike the legacy signed score, this is invariant to variant ordering, but it loses the distinction between *decline* and *increase*; report signed contrasts separately against a **semantically defined** no-cue anchor.

5. `P_gold`: restricted answer-set normalized probability when a defensible set of answer alternatives exists; compare with **first-answer-token** and answer-token-chain scores, fully acknowledging closed-set conditioning.
6. `R_alt`: log-odds `log p(y* | x,q) - log p(y_decoy | x,q)` for independently prespecified distractor-derived candidate answers *when they are well-formed under the same unit/format*. This identifies answer competition more directly than a raw gold-string likelihood, but is not defined for every cue type.
7. `C_cal`: confidence calibration against *actual correctness* of predictions; do not call the teacher-forced gold-string log-probability an empirically calibrated belief in correctness.
8. `G_mech`: a causal output contrast on a matched clean/corrupt pair, ideally a logit- or probability-margin between predeclared physically relevant and distractor-derived alternatives, assessed jointly with free-generation outcomes.
9. `I_select`: a selective intervention's effect on nuisance sensitivity minus its effect on genuinely relevant information, answer accuracy and unrelated tasks.

Primary scientific distinction. An ideal symbolic solver yields the same *answer value* across irrelevant edits. It need not be modeled as assigning exactly equal probability to a fixed byte string under differently worded contexts. Probability invariance is a **desirable robustness convention**, not entailed by the mathematical certificate. The new study must test whether different notions of instability cohere empirically, rather than treating one as a theorem-induced truth.

4.1 Competing mechanistic hypotheses: predictions and falsifiers

Code	Explanatory hypothesis	Distinct prediction	Defeating result
H0-F	Answer-format / teacher-forcing artifact	Effects vary strongly with canonical rendering, prefixed response and answer tokenization; little or no counterpart in natural answers	Similar cue effects across controlled answer formats and in generated answer competition.
H0-U	Generic uncertainty / difficulty	Correctness, entropy and prompt complexity explain label; perturbing many nonnumeric irrelevant statements has similar effect	Factorially isolated numeric and causal-specific effects after matching these signals.
H1-C	Numeral-copying / lexical priming	Cue numeral representation flows toward answer token/logits, effects depend on exact digits/token overlap even if units are incompatible	Robust effects require semantic/physical relevance and survive numeral string mismatch.
H1-M	Magnitude/units competition	Near-equal magnitudes and compatible units jointly predict failures; cue representations affect intermediate magnitude comparisons	Effects entirely explained by raw token overlap after token-balanced tests.
H1-R	Relevance-gating failure	Explicit isolation/relevance structure controls whether a distractor is used; QK attention to relevance language or gated OV pathway	Gate manipulations do not change causal flow once surface form is controlled.

Code	Explanatory hypothesis	Distinct prediction	Defeating result
H1-G	General context routing	Cue information flows through broadly shared circuitry, but no distinct physics-specific function	A tight, physics-specific pathway proves necessary across held-out tasks.
H1-H	Heterogeneous mechanisms	Cues A, B and C have distinct behavior/readouts/components; apparent pooled effect is composition	Independently confirmed shared pathway with strong out-of-type transfer.
H2-P	Specific partially recoverable pathway	Predeclared upstream sources, edges, head output and downstream targets pass patch/knockout/rescue on unseen clusters	Only isolated head ablation changes scalar sensitivity; proposed path does not mediate outcome.
H3-S	Selective repair	Mechanism-informed intervention improves cue invariance and accuracy with small damage and outperforms cheap baselines	Improvement is cosmetic, not generalizable, or is dominated by prompt removal.

High-priority prediction. Distinguish numeral copying from physically meaningful competition using orthogonal manipulations of numeric closeness, digit overlap, unit compatibility, stated isolation, task relevance and output format. Without this experiment family, a 'physical reasoning mechanism' claim remains underdetermined.

4.2 Claim hierarchy and evidence gates

- **C1 (benchmark):** certified task-answer invariance **plus** audited natural-language fidelity, independence, coverage, and well-specified exclusions.
- **C2 (behavioral):** reproducible cue-specific *behavioral* instability on unseen template clusters, reported across multiple endpoints rather than one answer-string likelihood.
- **C3 (monitor):** correctly scoped single-input monitoring with meaningful *paired* incremental value beyond tuned available baselines and enough precision to resolve deployment tradeoffs.
- **C4 (causal localization):** multiple valid intervention families identify specific components, replicated on a held-out causal panel and not explained by model damage.
- **C5 (mechanistic explanation):** a concrete, falsifiable functional account with necessity + controlled sufficiency + upstream/downstream path tests + causal abstraction/generalization.
- **C6 (repair):** selective, generalizing intervention outperforming appropriate nonmechanistic alternatives with a documented utility/cost frontier.

Promotion rule. A claim only advances one level when the corresponding tests pass; a highly significant C3 monitor does not establish C4/C5. The next paper needs either a convincing C5, or an exceptionally well-validated C1–C3 methodological finding with true novelty beyond current benchmark literature, preferably C6 if feasible.

5 Ordered execution: the twelve critical-path blocks

Revised dependency (Phase B is mandatory). B0 runs alongside P0; B1 and P1 reuse the same frozen DECOY-v1 audit; B2/B3 create the formal behavior hypothesis and the small v2 discovery grammar; B4 validates the semantic contract; B5 pilots independently sourced external adaptation; B6 freezes the behavioral signal and confirms it on structures that were never used in generator/probe/head selection; B7 approves the benchmark for its limited intended uses. **Only then** scale

P2/P3 and run confirmatory P4. P5 (monitoring) and P6–P8 (mechanism) can proceed in parallel only after Phase-B gating and P4 freeze trustworthy partitions. Exploratory H11 checks may happen earlier if marked *exploratory and nonclaim-bearing*. P9 (external replication) begins after a causal hypothesis is frozen. P10 (repair) begins after P7 yields a specific intervention. P11 (paper synthesis) continuously updates a claim ledger but freezes final claims only when the required branches close. Avoid a calendar-driven plan: each gate is about correctness, not arbitrary dates.

5.1 P0. Independent evidence reproduction, release provenance and code audit

Question. Which of Phase I's reported numbers actually follow from exact datasets, versions, model weights and scoring code? This is not clerical. A corrupted label, leaked fold or incorrectly aligned hook invalidates downstream mechanistic interpretations.

P0.1 Build an immutable artifact registry. Assign each canonical family and numeric instantiation UUID; freeze hashes of symbolic template, rendered text, response prefix, reference answer, model revision, tokenizer, generation configuration, scoring script, fold ID, monitor checkpoint, donor/target identities, intervention site, scheduler job, git SHA and random seed. Associate one machine-readable `claim_id -> exact artifact paths + generator code + checksum + eligibility rule + source doc table/figure`. Mark exploration vs internal confirmation vs truly untouched confirmation. Include missing, cancelled and zero-row runs.

P0.2 Reproduce submission tables and plots bottom-up. Starting from per-rendering records (not summary CSVs), regenerate 155→140→135 inclusion logic; every cue/domain rate; Qwen/Llama/DeepSeek label overlaps; AUROC/AUPRC, Brier, ECE, FNR@FPR; the leave-one-family-out out-of-fold predictions; paired 10,000-bootstrap *distributions*, not merely intervals; Qwen layer-16 head subset ranking; causal run per-family numerator/denominator and CI; 15-family transfer membership. Never silently fill missing records. Compare raw generated answers to parser canonicalization and teacher-forced input IDs.

P0.3 Re-run a small random selection end-to-end. Preselect at least 10 families stratified by model, label, cue type, gold-token length and reported effect, including CM_B_UM_059, plus five adversarial edge cases (near-match, Cue C, negative recovery, parse failure, one zero-like answer). Require exact determinism of frozen logits within a justified tolerance and consistency between raw hook-based and reference forward passes.

P0.4 Inspect the code for leakage and cohort selection. Verify scaler/feature construction, hyperparameter selection, threshold selection, layer selection, calibration, stacked predictors and correctness residualization are *inside* the appropriate nested training loop. Inspect whether first discovered candidate representations were selected using future confirmation families, whether class labels use model-generated parsing as an inclusion filter, and whether the same or structurally related templates cross the training/test border. Conduct an adversarial code review with a second investigator tracing one train/test observation by hand.

P0.5 Fix provenance discrepancies explicitly. The submitted manuscript says renamed-donor controls had **zero evaluable rows**; the older Part-II record says these controls were completed with zero recovery. Resolve from job logs and donor-selection code, then run the corrected control only after freezing the eligible target/donor populations. The later Llama head-2 sweep (~1.85 in the appendix) is **exploratory with incomplete provenance**; do not upgrade it without a complete clean replication.

Deliverables. `evidence_registry.parquet`, machine-readable claim map, frozen manifests, recomputation notebooks/tests, inclusion-flow report, code-review checklist, reproducibility docker/conda lock, discrepancy log including unfixed defects. All values from the paper should be labeled `verified`, `different but explained`, `unreproducible`, or `not enough source data`.

Gate P0 (hard stop). Proceed only if the behavioral likelihoods and labels can be independently regenerated and the intervention hooks demonstrably affect the claimed tensors and token positions. If any headline changes materially: publish a versioned erratum in the internal ledger, invalidate derived findings, and rerun the minimal upstream chain. Never patch new experiments onto a broken denominator.

5.2 P1. Measurement validity: challenge the entire sensitivity construct

Question. Does the observed signal reflect a model's improper computation or mainly a property of the string and scoring interface? A robust measurement paper would be valuable even if the original construct fails.

P1.1 Anchor-free full-factorial rescoring on the existing 140 families. From the four stored raw gold log-probabilities, compute: legacy `max_{j != base}(L_base-L_j)`; range `max(L)-min(L)`; SD, MAD and median absolute pairwise shift; pairwise signed differences; signed contrasts against a *new no-cue rendering*. Randomly permute the designated base through all four values **without rerunning the model** and report label stability, confusion matrix, reclassification rate, impact by cue type, and whether model differences preserve sign. Document that base-free range is always nonnegative and therefore has a different null distribution from the signed legacy statistic.

P1.2 No-cue and nonnumeric controls. For each sampled family, generate (i) cue absent, (ii) irrelevant numeric distractor, (iii) irrelevant nonnumeric sentence of comparable length, (iv) unrelated numeric metadata, (v) physically relevant number edit that deliberately changes the correct answer and is separately re-certified. Compare effects as matched differences-in-differences. This disambiguates sensitivity to the edit from generic extra information and proves the assay responds to genuinely relevant information.

P1.3 Output-wrapper multiverse. Score the same *semantic answer* under exact frozen wrappers: plain digits; `Answer: <value> <unit>`; boxed LaTeX; standardized JSON scalar; natural-language short answer; fixed number of decimals and scientific notation. Preserve units correctly; for incomparable strings, analyse within-wrapper contrasts rather than averaging raw log-probs. Compare model chat templates, system instruction, generation header, assistant-prefix token span and explicit stop tokens. Record exact tokenizer IDs for every answer sequence, correct prefix and target span; verify no accidental inclusion of prompt or padding tokens in the scored log-likelihood.

P1.4 Length and tokenization controls. For each family, compute total and per-token gold log-likelihood; answer-token count; answer string length; number of numeric subword units; target unit-token rarity; likelihood on numeric value alone vs value+unit; first-token vs full-sequence sensitivity. Stratify within target-token-length bins, match families on exact length where possible, and rerun with equivalent canonical answers differing in the number of subword tokens. Test whether the 0.5-nat legacy label is systematically easier to cross for longer answer sequences.

P1.5 Proper likelihood comparisons. Where a finite, solver-generated set of distractor-derived candidate answers is meaningful, compute a normalized restricted-choice distribution over `y*`, `y_decoy`, plausible arithmetic alternatives and `none-of-the-above`. Verify that candidate strings are matched for output grammar/units and quantify sensitivity to distractor-set construction. Report correct-vs-decoy log-odds (answer competition) alongside raw absolute gold probability. Separately, calculate semantic answer distributions from repeated free generations at several temperatures, clustering equivalent answers; compare with teacher-forced endpoints without pretending that finite-sample JSD is exact.

P1.6 Free-generation experiment. For matched cues, generate answers under deterministic decode and repeated stochastic decodes (e.g., three temperature settings, 5–20 samples per cell where budget permits), using a locked, audited parser plus blinded human checks on an error-enriched sample. Separate `correct`→`correct with confidence shift`, `correct`→`wrong`, `wrong`→`correct`, `wrong`→`same wrong`, `wrong`→`different wrong`, invalid/abstention. Report whether cue effects change numerical final answers, not only gold log-prob. Include a model-evaluated explanation of whether the cue is relevant as a *correlate*, never ground truth for internal use.

P1.7 Test metric reliability. Re-score identical prompts across repeat passes, alternative precision/quantization modes and batch configurations; for deterministic forward pass, require near-exact agreement. Test text-only punctuation, whitespace, equivalent mathematical unit renderings, equivalent output display precision, and instructions that change no physics. Compare measured cue shift to this baseline 'presentation noise' distribution.

P1.8 Separate output-level invariance from decision-level reliability. Evaluate selective prediction: confidence bins for final-answer correctness, ECE/Brier with family-cluster confidence intervals, and a contrast between stable/unstable families at matched difficulty. A raw fall in p(y*) may be a* desirable uncertainty response* if the cue makes the question harder to parse; the practical risk is overconfidence, erroneous answers, or unexplained drift despite certified invariance. Report each explicitly.

Design. First run P1 on a balanced pre-registered **audit subset** of original families; then a full frozen rescore on all eligible originals. Use source family as inference unit, model as repeated measure, and nested variance components for multiple wrappers/cue variants. Register a small family of *primary endpoints* (reference-free range, free-generation correctness transition, and correct-vs-decoy log-odds where defined); all other tests are sensitivity analyses.

Gate P1. If the reference-free, parser-independent phenomenon is stable and materially exceeds matched formatting/no-cue effects, proceed to new benchmark creation. If not: stop calling the current S_lp a shortcut construct; redirect to a scientifically valuable study of **how answer-string scoring creates apparent counterfactual fragility**. If free-generation remains invariant while teacher-forced p(y*) drifts, restrict the thesis to likelihood sensitivity, with no safety/correctness claims. If only Cue C exhibits enormous effects, separate it entirely as a format/unit-conversion phenomenon.

5.3 P2. Scale the Phase-B-approved DECOY-v2 design (not the old benchmark)

Relationship to Phase B. B2–B6 govern *whether*, *what*, and *how much* new data should be built. P2 is the implementation and capacity-expansion package after those approvals; it must reuse the validated graph schema, factorial feasibility matrix, accepted scoring endpoints, and locked grouping policy. Do not create 300–600 examples to make the dataset look large if the B6 pilot fails. Old Cue C is analyzed separately in V2-B, not pooled with V2-A.

Question. Can we independently manipulate the *proposed mechanism's causal ingredients* while preserving physics, and hold out genuinely new structures?

P2.1 Define an explicit generative grammar. Build symbolic programs with typed quantities, dimensions, causal roles and assumptions. Make the equation a directed computation graph with nodes for governing inputs, intermediate quantities and the target. Every non-governing distractor must be accompanied by a proof/certificate of nondependence and an **independent semantic checklist** that verifies the rendered text matches the graph. In positive-control families, vary genuinely relevant variables and recompute y* exactly. In 'mis-specified' negative-control families, deliberately insert a subtle physical coupling and confirm validators/verifier reject it. Do not trust free-symbol absence alone if the omitted quantity changes assumptions, reference frames or boundary conditions.

P2.2 Factorial cue design. Cross six primary axes while managing combinatorial explosion through prespecified orthogonal/blocked or fractional-factorial designs:

- **Numerical closeness:** exact-match, near (predefined log-relative distance bands), moderate, far; negative values where physically meaningful; *match numerical closeness while breaking digit overlap* and *match digit overlap while breaking numerical closeness*.
- **Physical status:** relevant variable, irrelevant isolated variable, same-unit non-governing quantity, dimensionally incompatible number, redundant-but-physically-equivalent expression, inconsistent/contradictory statement excluded from invariant-family claims.
- **Units/dimensions:** exactly same units, convertible compatible units, unitless, incompatible, composite units, symbol-only numeric distractor, scientific notation; verify conversion factors rather than assuming textual equivalence.
- **Semantic relationship:** explicit mechanically isolated subsystem, spatially separated system with a stated no-interaction assumption, same system but non-governing metadata, decoy framed as possible answer, decoy as historical/quoted example, redundant true fact.
- **Placement/format:** before/after governing variables, adjacent to the question vs far from it, parenthetical vs prose vs table, cue at first/middle/last segment, keyword emphasis vs neutral, one vs multiple occurrences; matched length and answer position.

- **Task structure:** one-step vs multi-step equation, intermediate near-match vs final-answer near-match, algebraic vs numerical answer, inverse problem, single vs multiple constraints, familiar vs newly composed lawful systems.

P2.3 Crucial deconfounding cells. For every target measurement identify a cell with equal cue-digit tokens but different physical relevance; equal physical relevance but different digit overlap; matched target-answer value but different governing law; same law with different numerical instantiation; same numeric cue and units but different isolation sentence. This creates counterfactual tests that a simplistic lexical or generic difficulty explanation cannot simultaneously satisfy.

P2.4 Quantity and independence. Target **the number of truly distinct symbolic structures justified by Phase-B pilot power, validator capacity, and lawful variation**, not merely several hundred paraphrases of 34 clusters. Retain 300–600 independently authored/certified new canonical structures with 4–8 cue conditions and 2–3 numeric instantiations *only as a provisional planning ceiling/range subject to B6*, not an acquisition quota or guaranteed power. Reserve complete template clusters, governing-law combinations, and source collections for locked confirmation; count them, not just surface renderings. If the number of genuinely different physical computation graphs is too small, prefer fewer valid distinct families and be candid about scope rather than relabeling siblings as independent observations.

P2.5 Orthogonal physics extension. Keep classical mechanics and basic circuits as core; expand deliberately to dimensional-analysis problems, waves/oscillations, optics, energy conservation, and selected thermodynamics *only when assumptions and equilibrium/path conditions can be exhaustively certified*. Place each new domain into a separate external cohort rather than pooling it into a favorable aggregate. Physical domain difficulty is a measured covariate, not a reason to exclude hard results.

P2.6 Out-of-distribution sources. Include human-written problems from newly devised curricula; naturally sourced textbook/exam-style material with licensing and contamination-risk audit; algorithmically synthesized nonstandard legal setups; new symbolic compositions unconnected to DECOY. Use source-held-out and law-held-out tests. Avoid claiming no training contamination in black-box base-model pretraining; instead test memorization proxies, novelty, and paraphrase sensitivity.

P2.7 Synthetic but adversarially plausible. Ask independent LLMs to identify wording ambiguity and suggest semantically tricky distractors, but *not to assign physics ground truth*. Require expert rewriting and symbolic validation before admission. Screen cases where the irrelevant cue makes the natural language pragmatically ambiguous even if the authors' algebra says otherwise. Publish rejection reasons and excluded families, not just surviving clean positives.

P2.8 Matched negative and positive sets. No cue; stylistically matched irrelevant text; numeric cue with wrong units; cue relevant by changing the actual physical dependence; answer changed in a certified and controlled way; solver question switched to a different target that makes the previously irrelevant variable relevant. This last *role reversal* is essential to demonstrate a selective relevance mechanism rather than indiscriminate digit suppression.

P2.9 Domain-specific robustness. Check exact dimension conversions, alternate significant-figure conventions, sign and frame transformations, force conventions, rounding, algebraic simplification, near-zero denominators and degenerate limits. Anything requiring tacit real-world assumptions receives an explicit stated-assumption audit and may be excluded from the core.

Gate P2. New benchmark admission requires verifier pass, independent human semantic pass, no open leakage collision, and a declared structural parent/cluster. If balanced factorial creation produces obvious artificial text, run a blind naturalness comparison against textbook-like prompts and improve the grammar; do not keep artificial examples because their effects are large. If independent law breadth cannot be achieved, reduce the claimed generalization scope and increase mechanistic depth in a narrower subdomain.

5.4 P3. Scale independent expert validation, parser reliability and leakage firewall

Relationship to Phase B. The validator qualification/pilot, v1 physical-semantic audit and external adaptation viability are already required in B4/B5. P3 delivers full new-data validation, blinded auditing of the locked confirmation split, response-parser adjudication, and cluster-separation audits. A positive model result cannot waive these requirements.

P3.1 Personnel and blinding. Recruit **two independent physics-qualified validators per confirmatory family**, with a third adjudicator; calibrate on easy, adversarial and genuinely ambiguous cases. Validators must not see model outputs, sensitivity labels, probe scores, or the hypothesis that generated the family. Ask separately: equation correctness, sufficient assumptions, cue irrelevance, answer invariance, naturalness, potential missing interactions, unit correctness, correct rounding, and whether paired renderings differ only as intended. Record annotator confidence and reasons.

P3.2 Quantify disagreements. Predefine exclusion and adjudication rules. Report raw agreement, Cohen's kappa for core binary properties plus prevalence-aware and/or Gwet agreement if severe imbalance makes kappa misleading, bootstrap CIs at structure level, domain-specific failure rates, adjudication edits and changed-label counts. A 0.80 kappa aspiration is not a license to drop difficult cases to improve agreement.

P3.3 Mutational verification. Apply deliberate failure mutations: cue leaks into equation; change an assumption; remove mechanical isolation; unit conversion error; sign flip; change relevant variable; rename cue to relevant variable; swap target quantity; add a hidden dependence; omit a necessary boundary condition. Include algebraically equivalent positive mutations. Report sensitivity and specificity of the *validator/verifier system* against the **known mutations**, stratified by mutator, and verify that invalid examples fail for the intended reason.

P3.4 Parser study. Draw a blinded, stratified sample of responses across models, cue types and generated formats including refusals, multiple numbers, intermediate answers, hallucinated units, LaTeX, scientific notation, significant figures, symbolic answers and truncations. Require two human adjudicators. Report coverage, error rate, coverage conditional on outcome, answer-equivalence false positives, and impact of parser uncertainty on behavioral rates via conservative bounds. Keep ambiguous responses as an explicit category.

P3.5 Multi-level overlap detection. Build union-find clusters from exact prompt hashes, number-masked text, normalized symbolic AST, equation graph isomorphism, governing-law family, tokenized frame, variable-renamed graph, cue-grammar identity, near-duplicate embeddings, manually reviewed semantic relationships, and identical solution programs. Freeze a transparent clustering policy and manually adjudicate a representative sample of both 'same' and 'different'. `Same law` is not always leakage, but should be explicitly held out in a stricter transfer test. Group by connected components of the *relevant* relation so transitive siblings never straddle splits.

P3.6 Audit the original 34 clusters. Review the 2,352 unresolved rendering pairs and 147 family pairs; do not automatically treat them all as leakage or as harmless. Re-evaluate the original LOOCV by cluster-held-out and by distinct symbolic program. Report overlap with new locked confirmation; newly generated semantic paraphrases of old computation graphs remain in discovery, not independent replication.

P3.7 Freeze cohort version. Manifest the discovery set, internal confirmation, external confirmation, adversarial challenge and human-written source set. Every family and variant receives an immutable split assignment before labels are evaluated. A family removed after outcomes are seen retains a tombstone with reason; report intent-to-evaluate alongside eligible results. Keep a shadow test set inaccessible to all tuning scripts and agents.

Gate P3. No confirmatory hidden-state or causal generalization claim until each confirmatory family's natural language has passed the independent validation gate, and leakage risk is audited. If the certificate proves authors' equations but validators reject a sizeable fraction of prompt semantics, revise the benchmark and *restart* affected behavioral measurements; it is not a statistical nuisance covariate.

5.5 P4. Definitive behavioral mechanism discrimination and power planning

P4.1 Preregister a small core experiment. Primary contrast: interaction of numerical proximity × unit compatibility × relevance status on (i) free-generation answer contamination, (ii) correct-vs-decoy log-odds, and (iii) reference-free gold-likelihood dispersion, with distinct hypotheses about each. Use held-out structural clusters and within-structure matched randomization. Prespecify which cue, output protocol and model form the headline analysis, leaving broad multiverse tables as secondary.

P4.2 Estimate detection power honestly. From a discovery-only pilot, simulate data with hierarchical cluster dependence, label prevalence and within-family correlation; evaluate width and coverage of CIs for interaction coefficients and paired AUROC differences at realistic effect sizes. Pick a confirmatory sample large enough for the *smallest scientifically meaningful effect*, not for detecting any departure from zero. Record attrition assumptions from parser/validator failures. If model-time limits prevent a narrow enough CI, make a precision-limited conclusion, not a powered claim.

P4.3 Multi-level regression. Fit transparent hierarchical mixed models or carefully predeclared GEE/cluster-robust models with structure random intercepts and, where estimable, random cue slopes. Test predeclared interactions (near-match × unit compatibility, proximity × digit overlap, relevance × isolation language, cue type × architecture, position × distance to answer). Avoid post-hoc feature selection across a massive factorial without familywise or FDR correction and confirmatory replication.

P4.4 Nonlinear near-match profile. Treat the current Pearson 0.074 as a warning. Compare a smooth monotone magnitude-distance model, a predeclared near-match threshold/tail model, and a token-overlap model. Use out-of-cluster held-out log predictive density / calibration rather than celebrating maximum in-sample correlation. Create *counterexamples* to each model (numerically close but digit-disjoint; digit-similar but far; exact numeric answer but incompatible units).

P4.5 Cross-model same-panel design. Use at least two independently implemented dense model families on the identical certified panels; add an independently selected reasoning-tuned model only with a verified reasoning/answer scoring protocol. Match output grammar without assuming model chat templates are interchangeable. Analyze paired family × model interactions. Document exact models, versions and instruction tuning, not just brand labels.

P4.6 Severity and validity stratification. Separate family-positive rate, median absolute effect, 90th/95th tail effects, answer flips, correctness transitions, high-confidence errors, model abstention, symbolic answer/rounded answer cases, and variants whose textual output is unrelated to gold. Verify effects persist when excluding the 1–3 most extreme family outliers. Report both means and robust quantiles.

P4.7 Core negative tests. If the alleged near-match effect is reproduced by an irrelevant bibliographic numeral; if unit incompatibility does not change effect; if 'mechanical isolation' does nothing; if the effect vanishes under mathematically equivalent output formats; if only lexical token overlap matters; or if generated answers stay invariant with no decoy competition, then reject the physical-relevance-gating thesis. Those are scientific results, not failures to hide.

Gate P4. Mechanistic discovery proceeds under a **specified winning behavioral contrast**. Examples: `near-match unit-compatible irrelevant cue changes gold/decoy log-odds but does not affect correct answer`; or `cue changes actual wrong-answer selection`. The contrast determines the mechanistic metric; do *not* reuse the legacy normalized recovery for all questions. If the result is only format-sensitive confidence drift, publish Track E or pivot to model-calibration analysis. If distinct cue types behave differently, choose one primary cue for deep mechanism and use others as explicit external scope tests.

5.6 P5. A definitive monitoring study: does access to hidden states actually help?

Thesis. A monitor can only be called an operational advance if it is deployable under the claimed input budget, calibrated for the chosen target, statistically and practically better than appropriate cheaper alternatives on unseen structures, and robust to shift. A model that reads four renderings is a *counterfactual-family diagnostic*, not a one-input detector.

P5.1 Define three separate prediction tasks. Task S (single-prompt): from one prompt's state, predict expected vulnerability to future *unseen* nuisance edits of that prompt; no other family member may enter the features at inference. **Task F (four-input):** given a full family, predict observed sensitivity from averaged or variance features; compare against a *four-query* behavioral measurement, not a one-query baseline. **Task V (variant-local):** predict the effect of a particular cue value or proximity state on a *particular* rendering; train on per-variant labels obtained by leave-one-variant-out contrasts with a semantically meaningful anchor, not merely copies of the family label. No pooled AUROC across these tasks.

P5.2 Freeze three primary feature sets. Single final-prompt residual stream at one predeclared layer from discovery; first/embedding layer as lexical control; next-token logits/entropy as cost-matched comparator. Additional sites (cue token, answer-elicitation prefix, attention output, MLP output, generation boundary) are discovery scans requiring a separate confirmatory test. Pooling means, variances, PCA or learned aggregators over multiple renderings is reported with actual query count.

P5.3 Establish a serious comparator ladder:

1. Untrained constant and empirical cue/law/base-rate predictors, plus difficulty-only model.
2. Prompt-only TF–IDF, character/word n-grams, digit features, physical units, parser/tokenizer statistics, length and structural AST features; identical CV effort.
3. Text semantic embeddings from a separately selected frozen encoder or the same model's early layer; probe complexity matched as closely as feasible.
4. One-pass output entropy, top-1 vs top-2 margin, reference-answer token likelihood (when answer known), energy/entropy across target vocabulary, logit norm and simple measures of syntactic uncertainty, all **fitted** inside nested training folds.
5. Generic correct-answer predictor, semantic-entropy probe if the required generation setting exists, and a stronger uncertainty representation/stability comparator adapted from published work; avoid claiming 'cost matched' if the comparator requires extra generations.
6. Black-box two-query and multi-query counterfactual consistency; direct answer uncertainty and sampled self-consistency; report real latency/GPU pass count.
7. CoT/rationale classifier, small local judge, larger independent judge, outcome/step critic where legally available; these use generated output and are not pre-generation competitors.
8. Fully tuned *combined non-activation* model, with proper nested stacking, calibration and feature selection. Hybrid with hidden states is only evaluated using out-of-fold training scores, never labels/features learned on the outer test family.

P5.4 Probe family and capacity test. Compare L2 logistic/regression, sparse L1/elastic-net, low-rank supervised projection with nested rank choice, small MLP with strictly limited capacity, fixed embedding cosine/margin features, linear residualized probe, and representational distance to class prototypes. Match or report trainable parameter counts and label requirements. Use random-label and label-shuffle controls, a *selectivity gap* relative to lexical controls, and two seed/fold repetitions. Do not automatically pick a nonlinear high-capacity probe if it overfits a small set of templates.

P5.5 Within- vs between-structure information. Predict across numeric instantiations of the *same* symbolic structure, across entirely held-out template clusters, across different governing laws, across new human-written law compositions, across held-out cue types, and across model checkpoints. If high AUC only appears when the train set contains the same governing law/templating scaffold, describe the signal as *template family recognition*, not general hidden-state vulnerability prediction.

P5.6 Negative controls against information leakage. Train on different random assignments of equivalent semantic answer strings; target permuted within structural strata; predict deliberately irrelevant metadata labels; train test separated temporally and by generative template source; evaluate a prompt-only syntax parser with numeric-structure features. Check that normalizing by answer token count, excluding Cue C, and no-cue-minus-cue contrast do not eliminate the nominal signal.

P5.7 Correctness and entropy deconfounding. Use predictive models trained out-of-fold on correctness/entropy, then estimate the *incremental* value of hidden-state scores via nested stacking and

paired loss difference on the external set. Do not residualize on test labels. Compare `entropy only`, `correctness only`, `entropy+correctness`, `surface+entropy+correctness`, `hidden only`, and `all+hidden`. Report change in AUROC, log loss, Brier, decision cost, ECE/AUPRC and confidence intervals; an AUC delta of 0.03 is not a result if the CI crosses zero.

P5.8 Model selection accounting. The original best L18 came from a 28-layer sweep, so account for the winner's curse. Choose a layer from a discovery set, freeze before external evaluation, and use max-T or hierarchical/false-discovery correction for full exploratory scans. Use pre-specified L16/L18 hypotheses only when not chosen retrospectively on the same confirmatory data.

P5.9 Calibration and policy value. Fit calibrators solely on the training/development groups (Platt, isotonic, temperature, beta calibration); choose one by validation only. Compare class-conditional reliability, worst-bin and uncertainty, calibration under cue shift and new domains, expected cost at explicit FP/FN tradeoffs and intervention trigger budgets. Report FNR at FPR=0.05 and 0.10 and an entire ROC/PR curve with CIs rather than cherry-picked operating points. Evaluate prevalence shift separately because precision and calibration can change as a function of positives' base rate.

P5.10 Informational upper and lower bounds. Report how much one model pass can predict of *counterfactual family vulnerability* when the relevant future cue edits are not yet sampled. Consider oracle features using actual future cue values and oracle family statistics as **nondeployable upper bounds**, making clear which prediction problem they solve. Evaluate whether a single-input feature can possibly predict a near-match distractor introduced only *later*, a potentially fundamental information bottleneck.

P5.11 Cost accounting. Time one forward pass, activation extraction, copying to CPU, classifier inference, output entropy and multi-query black-box checks on the same hardware and batch sizes. Provide memory bytes and wall clock per decision with uncertainty and practical batch throughput. A hidden-state monitor that costs the same forward pass but adds intrusive hooks may not beat reading next-token entropy in deployment.

Gate P5. Promote a monitor as a result only if a frozen **single-input** specification achieves a practically meaningful and statistically supported incremental advantage on a held-out structural cohort under a preregistered endpoint. If not: retain monitorability as a descriptive diagnostic, report null incremental value honestly, and redirect novelty toward mechanism (P6–P9) or benchmark measurement (Track E). An AUROC higher than 0.5 is not a substitute for incremental value.

5.7 P6. Re-localize the phenomenon under valid causal metrics and selection rules

Question. Is Qwen layer-16 head 11 a causal component of a *physical computation*, or does indiscriminate perturbation of its output merely change logit scales and the legacy ratio metric?

P6.1 Choose clean/corrupt pairs prospectively. For each problem structure prepare aligned no-cue, irrelevant cue, decoy near-match, decoy far-match, genuinely relevant edit and semantic-control prompts. Label pairs by what *changes in model behavior*, not by their inclusion in a high-S_likelihood set. Explicitly stratify `correct/wrong`, `logprob-only`, `answer-flip`, `high-confidence wrong`, and `high-S legacy but low reference-free`. Prefer same-token-count or equal-length spans within each model; when alignment impossible, patch semantically matched positions only and validate with position-ablation controls.

P6.2 Freeze the causal response variable. Primary: absolute change in `R_alt` or correct-answer logit/logprob margin for a paired, fully specified candidate set. Secondary: reference-free family dispersion, direct free-generation correctness change, KL divergence of answer distribution, and gold likelihood. Always report original score, intervened score, absolute difference, and denominator before any normalized recovery. Never average unstable `effect / near-zero original` ratios without thresholded eligibility and robust alternatives; clipped/uncapped summaries must appear together.

P6.3 Reproduce the Qwen effect as a new experiment. Use untouched held-out structures with prespecified L16 H11, the old four-head set, L15 and L17 flanks, near layers, unrelated heads matched in norm/attention entropy, and random-direction / random-site controls. Repeat original high-S panel

separately and document its selection. Report per-family scatter (before/after), absolute shifts and directional restoration; if a knockout *inverts* the signed drop, ask whether it changed base and cue likelihoods in opposite directions.

P6.4 Test several ablation semantics. Zero ablation, mean replacement with unconditional activation, corruption-based replacement, no-cue donor activation, same-answer irrelevant-cue donor, matched random vector, orthogonal projection/linear subspace removal, feature-level clamping and multiplicative rescaling. A zeroed head could force abnormal activations that trigger compensation; compare against on-manifold replacements and a path-specific natural contrast. Report a method-by-site stability table rather than interpreting one intervention flavor as definitive.

P6.5 Necessity must be specific. A candidate component is necessary *for the identified pathway on the tested distribution* if its controlled removal suppresses the chosen decoy-specific effect while preserving (within a preregistered tolerance) response to relevant input changes and unrelated problems. Test in- vs out-of-distribution target panels and under role reversal where the formerly irrelevant quantity becomes relevant. Avoid global claims of logical necessity: transformers may use redundant pathways.

P6.6 Sufficiency must use appropriate variables. Transplant candidate activation from near-match/sensitive to far-match/stable contexts (with clean shared physics), and vice versa; dose sweep both directions; test target logits *and* generated choices; compare content-preserving swaps from same-answer, same-law, same-unit but irrelevant donors. If direction injection changes the legacy S score but not answers or gold/decoy competition, do not call it sufficiency for shortcut use.

P6.7 Detect off-target damage and self-repair. Test ordinary accuracy on an independent control suite including no-cue physics, genuinely relevant parameter edits, unrelated numerical and lexical tasks, common natural-language QA and syntactic completion. Measure loss and KL distribution shifts, next-token entropy, output length, and follow-on compensatory changes in nearby heads/MLPs. Ablation can trigger backup mechanisms, as shown in prior work on the Hydra effect; test local replacement vs global zeroing and cross-layer adaptations.

P6.8 Revisit Llama and DeepSeek correctly. Recover original exploratory Llama L21 H2 candidates, implement clean provenance, validate sign and magnitude on a newly selected model-positive panel, and test predeclared layers/heads including matched negative controls. For DeepSeek independently confirm prompt boundary / reasoning trace / final-answer token scoring and match causal denominators before concluding it lacks the Qwen pathway. If a site misses, do a discovery sweep on an exploratory set, not repeated scans of confirmation families.

Gate P6. If the Qwen effect disappears under reference-free and correct/decoy metrics, retire the claim that H11 mediates physics shortcut sensitivity; treat it as an endpoint-specific sensitivity artifact and investigate why. If it survives only zero ablation but not mean/replacement, it is a *fragile lesion effect*, not a reliable circuit. If it survives all controls but not free-generation accuracy, call it a cause of likelihood/answer-competition change, not of success/failure in physics. Only a robust, specific effect advances to P7.

5.8 P7. Identify what the causal components actually compute

Question. What computable function connects distractor tokens to altered answer tendencies? Localization without function is insufficient for a high-impact mechanism claim.

P7.1 Anatomize head 11 at L16. For every token position (governing variables, cue value, cue unit, isolation/irrelevance phrase, problem question, last prompt position, assistant prefix), measure QK attention patterns, attention logits, output vector norm, OV map contribution and direct residual/logit attribution across controlled factorial pairs. Do not treat an attention heatmap as causal evidence. Patch *attention probabilities* and *value vectors* separately; patch source-token contributions into the target position, then test whether they mediate the response. Compare input-token-level Q/K/V sites and output-projection pathways with attention/MLP processing nearby.

P7.2 Test explicit candidate functions head-to-head.

- **F-copy:** retrieves the largest/nearest/recent distractor numeral or copies its digit prefix.
- **F-match:** encodes equality/proximity of numeral and target; may trigger when two quantities approximately coincide regardless of meaning.
- **F-unit:** recognizes dimension/units, especially numerical objects with units matching target.
- **F-role:** tests sentence-level semantic disconnection (mechanically isolated, separate system) and suppresses an otherwise attractive candidate.
- **F-position:** attends to near-query tokens or sentence order independent of meaning.
- **F-answer prior:** globally shifts confidence in one output format/answer length without representing the distractor.

For each function build at least **two interventions that predict different effects** from the alternatives: e.g. the same digits with wrong units (F-copy yes, F-unit no); same magnitude but digit-disjoint exponential form (F-match yes, F-copy no); same tokens moved elsewhere (F-position changes, F-match preserved); switch target to make cue relevant (F-role reverses, F-copy may not). Use a held-out grid, not anecdotes.

P7.3 Controlled activation decoding with interventions. Train simple nested linear probes from the *candidate causal head output* to predict cue numeral, target magnitude, digit overlap, unit compatibility, semantic isolation, relevant variables and a binary importance tag. Probe success is only a correlate; complement with subspace interventions removing **one** decoded variable while preserving others and testing actual behavior. Quantify whether causal directions for numeral/units/role are geometrically independent, oblique, or redundant. Compare causal component features with the L18 monitoring direction to test whether diagnosis and mechanism overlap.

P7.4 Representation geometry. Construct matched activation differences `delta(z_near,z_far)` and `delta(relevant,irrelevant)` at key sites. Quantify cross-structure alignment with CKA / linear CCA / principal angles / projection energy, controlling for tokenization and model width. Classify whether differences lie in a stable low-rank subspace or are high-rank context-specific. Fit subspaces **only on discovery** and predict held-out causal effect from projection magnitude. Include nearest-neighbor and random subspace nulls of equal dimension.

P7.5 Direct logit attribution. Analyze head and MLP contributions to specific pre-registered answer token logit differences, not merely full vocabulary entropy. Follow with true counterfactual patching because linear logit attribution can omit nonlinear downstream transformations. Decompose whether the head changes *gold score*, *decoy score*, both, or generic output length; report full effects with nonlinear residuals.

P7.6 Temporal localization at the prompt and generation boundary. Where token alignment is rigorous, score the first answer token and teacher-forced first intermediate physical quantity, then final answer. Separately follow independently generated explanations only via *semantic event alignment* (e.g. law chosen, intermediate quantity emitted) and mark such trajectory analyses descriptive unless teacher forcing or controlled resampling makes positions comparable. Re-run interventions only at prompt positions when causal token comparability fails.

P7.7 Invariance and role reversal. A legitimate relevance gate should ignore irrelevant `z` but respond when a modified problem makes `z` truly relevant. Patch the proposed representation between a cue-irrelevant and cue-relevant version with carefully matched numerical and syntactic content. Predict *both* the normal and role-reversal responses. This is more discriminating than simply making the model less sensitive to numbers.

P7.8 Hypothesis scorecard. Each candidate function has a predeclared sign prediction over the factorial cells, a held-out prediction score, selective knockout/patch effect, falsifying control, alternative it beats, and failure case. Prioritize the simplest function that explains *held-out intervention effects* over the one with most attractive activation visualizations.

Gate P7. Advance only if at least one explicit functional hypothesis predicts effect direction across independent manipulations and survives direct causal tests. Otherwise report a localized but **uninterpreted** causal handle, explore larger multi-component representations (P8), or pivot Track E.

Do not label an activation feature 'physics-relevance' merely because high-scoring examples sound physics-related.

5.9 P8. Recover and validate upstream/downstream computation, not just one head

P8.1 Define sources and sinks before attribution. Sources: cue numeric tokens, cue unit token, semantic isolation phrase, physically relevant quantities, governing-law words, output-format instruction. Candidate sinks: L16 H11 (or discovered stable mediator), L18 diagnostic state, correct-vs-decoy answer logit. Predefine the desired causal contrast: `near-match irrelevant cue` vs `far-match irrelevant cue` with shared target answer; add a *separate* relevant-edit contrast in which the gold answer legitimately changes.

P8.2 Two independent discovery procedures. Use gradient-based edge attribution/attribution patching (and integrated-gradient refinements when attribution saturates) as a **candidate ranking method**, plus targeted clean/corrupt activation patching at layer/position/head/MLP sites. Optionally compare circuit mask optimization. Causal confirmation uses true interventions, not approximate gradients; attribute graph methods can be misleading under reconstruction errors and feature approximation.

P8.3 Path patching. Rank upstream attention heads, MLP outputs and input-token contributions to the mediator. Apply receiver-conditioned path patching so a candidate source is changed while alternative branches are held fixed as far as the implementation permits. Repeat for source→mediator, mediator→downstream layer, and downstream→answer. Check token and hook order, causal graph direction, numerical tolerance and reverse interventions. Demonstrate that the identified path matters *only* when the mediator receives the source information.

P8.4 Knockout–rescue logic. (a) Knock out upstream source, measure lost effect; (b) rescue with the mediator's clean activation, test restoration; (c) knock out mediator, attempt upstream rescue and predict failure if mediator is necessary; (d) control with unrelated and matched same-answer donors; (e) rescue a role-reversal relevant-edit task to show the path is not simply a destructive filter. Contrast against rescuing a generic next-token loss change. This sequence is a stringent causal ordering test.

P8.5 Circuit completeness and minimality. For a hypothesized edge set, measure `faithfulness`: run the model retaining only hypothesized pathways versus replacing complementary pathways with matched control activations. Evaluate sufficiency, comprehensiveness, selectivity, stability across corruption choices and path sparsity. Generate faithfulness-vs-circuit-size curves and a specificity-vs-size frontier; report residual unexplained effect. Check that random matched graphs and a trivial high-capacity whole-residual patch do not provide the same explanatory power per component.

P8.6 Causal abstraction validation. Write an explicit high-level algorithm (e.g., parse relevant variables → parse candidate numeric values → evaluate relevance/unit consistency → choose candidate answer); define which neural subspaces map to each high-level variable. Use interchange interventions on paired input families and test whether the low-level model's responses match interventions predicted by the high-level algorithm. Vary one high-level variable at a time while holding correlated variables matched. Poor interchange agreement falsifies the proposed explanation even if component knockouts are large.

P8.7 Representational bottleneck hypotheses. Test whether the mediator stores raw value, normalized magnitude, a relevance flag, a source identifier, or a merged answer candidate. Probe separate source/value/role subspaces, conduct orthogonal projection ablations and donor swaps, quantify whether the same component acts as a routing gate or downstream logit amplifier. Confirm on entirely held-out laws, magnitudes, prompts and cue positions.

P8.8 Sparse features: optional, validation-heavy. If the low-rank/head-level account stalls, compare existing open SAEs, transcoders/cross-layer transcoders, supervised distributed alignment search and ordinary MLP neurons **on equal compute and feature budgets**. Evaluate reconstruction error, feature stability under seeds, causal interchange and suppression specificity; do not equate a semantically appealing SAE feature with a mechanism. MIB reports that SAEs are not always better than neuron baselines for causal variable localization; treat SAEs as a *method to test*, not a badge of depth.

P8.9 Synthetic ground truth. Train a small controlled transformer or fine-tune a small open model on synthetic physics-graph tasks with a deliberately planted spurious numeric rule or well-specified shortcut curriculum. Know exactly which cue correlation was taught; assess whether the discovery pipeline recovers the planted variables/path. Add matched models trained without the shortcut. Recovering a known route does not prove the same route operates naturally in Qwen, but failure to recover a simple planted route calls the method into question.

Gate P8. A mechanistic paper needs a compact algorithmic story with validated source-to-mediator-to-sink effects on untouched families and counterfactual predictions that a simpler lexical/generic alternative cannot explain. If multiple redundant routes exist, report a distributed or partially redundant mechanism; do not force a single-head narrative. If only correlation graphs emerge, limit conclusions to localization, not circuit recovery.

5.10 P9. Cross-model, cross-domain and post-training generalization

P9.1 Cross-architecture confirmation. Freeze the *function hypothesis*, not the head index. Use fresh discovery samples to localize functionally analogous components in a second architecture; evaluate the hypothesis on untouched clusters with matched causal endpoints. Distinguish transfer of a **causal variable/function** from transfer of individual hidden-state directions or identical layer depths. A functional analog in a different head is replication; a naive matched-layer knockout failing is not a complete null.

P9.2 Scaling ladder. Conditional on compute, evaluate small→medium→large checkpoints within a model family, avoiding a giant number of models that each receive superficial analysis. The core comparison should control architecture and training stage enough to interpret causal differences. Report behavior across scale before choosing expensive causal cases; instrument only models with stable access to required hooks and answer/trace scoring.

P9.3 Same-backbone post-training. Compare base/instruct and, where available, same-backbone reasoning fine-tunes, with changes to system and answer prompting controlled. Analyze whether post-training changes *rate of error*, *confidence sensitivity*, *identified causal function* and *component allocation*. Do not attribute differences to RL without isolating comparable training runs or stating that multiple post-training changes are confounded.

P9.4 Familiar-to-novel-law transfer. Create new physically valid symbolic worlds or dimensionally coherent toy laws with unfamiliar names/variables, teaching necessary rules explicitly. If a numeric shortcut generalizes to these tasks, it suggests a broader numerical/linguistic mechanism; if it disappears, familiar physics priors may be essential. Make sure the toy laws are learnable and that difficulty is matched by solver-assisted pilots.

P9.5 Source and language transfer. Test human-written problems, new problem formats, terse structured equations, narrative descriptions and controlled bilingual renderings where native/qualified validators can certify semantic equivalence. Record tokenization differences rather than confounding translation with mechanism. English-only results do not imply cross-lingual generality.

P9.6 Cue composition and adversarial boundary. Add 2–3 simultaneously irrelevant cues; vary distractor congruence/contradiction, near-match competition and ordering. Look for additive, superadditive or suppressive interaction in behavior and circuit effect. A mechanism that only explains single numerical cues must be scoped accordingly. Adversarially search for maximally sensitive *valid* held-out examples under a fixed query budget, then retest on freshly authored confirmation examples to control optimizer overfitting.

P9.7 Persistence across tools and precision. Use two independently implemented hook backends when feasible; compare fp16/bf16 and unquantized vs modest quantization (with a numerical agreement gate). If a component's effect depends on a software implementation bug, retire it. Cross-backend reproducibility is a stronger assurance than adding unrelated models.

Gate P9. Write separate conclusions for behavior, monitor, function and anatomy. If the function recurs across architectures but component identities differ, publish that heterogeneity as part of the mechanism. If only Qwen shows it, a deep Qwen case study is possible but needs stronger within-Qwen causal proof and a clear reason why the finding generalizes conceptually; do not state a universal 'LLM shortcut circuit'.

5.11 P10. Mechanism-guided repair and comparative efficacy

P10.1 Define operational loss. Optimize weighted free-generation error, cue-related answer-flips, correct/decoy log-odds, calibration and OOD robustness, subject to maximum allowable degradation on no-cue and relevant-variable tasks. Avoid optimizing only a teacher-forced gold-likelihood statistic, which could be improved by reducing likelihood everywhere or artificially flattening logits.

P10.2 Nonmechanistic baseline ladder. Remove the distractor with a deterministic semantic filter; prompt an explicit 'ignore irrelevant information' instruction; rephrase into governing variables; answer via a small symbolic solver; use two-query consistency checking; majority vote/self-consistency; targeted counterfactual augmentation/fine-tuning. Compare latency/compute and the damage from mistakenly ignoring relevant variables. This ladder sets the bar for any purportedly superior circuit edit.

P10.3 Training-free circuit interventions. Head-level mean/no-cue replacement conditioned on a high-specificity trigger; linearly project out a causal cue-direction; gate a *proven* source→mediator edge; apply tiny direction subtraction/addition at a specified site; selectively patch from a matched no-cue rendering. Tune intensity on development only, with a strength-vs-damage curve. Detect unintended damage on prompts where the same numeral becomes *relevant*.

P10.4 Monitoring-triggered repair. First measure the *actual single-prompt monitor's* precision and FPR under shift; test selective edit only for the triggered subset and compare with applying the same edit to everyone. Show deployment benefit under realistic base rates and decision costs. An oracle trigger based on four future variants is only a nondeployable upper bound.

P10.5 Training-based mitigation. If justified, train LoRA or lightweight fine-tunes using certificate-backed paired counterfactual consistency, plus a standard answer-correctness loss and relevance-preservation controls. Contrast with random distractor augmentation, strong-distractor curricula (cf. GSM-DC) and generic instruction tuning. Keep **entire** governing-law clusters out of training and compare relevant-variable responsiveness to avoid teaching the model to ignore all extra numbers. Follow training to see whether the original head mechanism weakens, relocates, or is bypassed.

P10.6 Stress test attack/defense equilibrium. Optimize valid cue substitutions, wording and placements to evade the monitor or defeat the intervention under a declared budget; report adaptation curves and fresh-attack transfer. Include adaptive attacker access variants (knowledge of threshold only, monitor score, full gradient) and certification of each generated edit. If the intervention only fixes in-distribution examples, classify it as narrow mechanistic control.

P10.7 Repair claims require causal linkage. Compare targeted edit, equal-norm random edit, nearby-layer edit, alternative-component edit, prompt-only baseline, and unedited. If repair improves outcomes but the learned mechanism does not explain **why**, report a useful empirical intervention without implying a causal circuit. A strong claim connects pathway suppression to counterfactual behavioral improvement and rescue/relapse.

Gate P10. A positive result requires held-out paired improvement in a practically meaningful outcome (preferably correctness/answer competition), statistically resolved harm bounds, and advantage or distinctive tradeoff against simple nonmechanistic methods. If basic cue removal works better and with less damage, do not pretend an expensive neural edit is operationally superior; it may still be a valuable mechanistic test.

5.12 P11. Confirmatory closure, adversarial review and publishable claim freeze

P11.1 Build the paper around one causal chain, not the chronological lab log. Figure 1: invariant-task causal factorial and hypotheses. Figure 2: measurement audit separating gold-string, restricted-choice and generated answers. Figure 3: cue-feature interaction on independent clusters. Figure 4: true patch/knockout/rescue and absolute effect sizes. Figure 5: upstream path with counterfactual predictions. Figure 6: replication and/or selective repair. Include failures in compact explicit tables, not buried in supplements.

P11.2 Mandatory final audit. Independent recomputation of main numbers, validation signatures, code and model versions, held-out contamination audit, pre-registration deviations, multiple-testing ledger, negative and zero-row results, exact intervention sites, confidence intervals, relevant baselines, alternative metrics, artifacts and statistical unit. Ask an unaffiliated research peer to attack the strongest claim with the raw evidence and with only the released protocol. Attempt to falsify before submission.

P11.3 Two-stage external confirmation. Internal hold-out is used for one explicit confirmatory decision. A fully untouched external cohort from newly written/law-held-out materials is opened **once** after hypotheses, functional representation, causal site and inference code are frozen. All additional tuning after opening the external cohort converts it into development; obtain new data for a subsequent confirmation.

P11.4 Reproducible release. Include rendered prompts, symbolic programs, typed equations, certificates, validator labels/adjudications, all eligible model outputs and log probabilities, inclusion metadata, per-family out-of-fold predictions, raw intervention before/after endpoints, code, environment, random seeds, hardware profiles and manifest checksums. Publish both passing and failing runs with an explicit authority hierarchy; include anonymized artifacts while under double-blind review where permitted.

P11.5 Make venue choice contingent on the result. An ICLR-level mechanism narrative needs a substantial **new insight**, not the most pages of supplemental ablations. If the core contribution becomes measurement science, consider an evaluation/benchmark framing. If the central advance is trained mitigation, frame the paper as robustness/causal control. Do not keep all three narratives at equal weight unless their evidence makes a coherent, noninflated causal chain.

Gate P11. No paper submission until an independent reader can reproduce the primary figures and state, in one paragraph, the causal explanation and the best attempt to falsify it. Disconfirming evidence must be integrated into the headline scope, not relegated to future work.

6 Complete downstream experimental/ablation inventory (registered candidates; Phase-B contrasts are additional)

This catalogue is deliberately broader than the critical path. Each code describes a **testable contrast**, not a vague recommendation to ‘try more experiments’. **R** = required on the critical path; **C** = conditional on earlier evidence; **X** = exploratory/high-upside. Selection of a confirmatory subset must happen before the corresponding test cohort is opened. A negative result removes the branch’s supported claim, not the datum.

6.1 A. Input and task semantics (A01–A18)

ID	Status	Contrast / causal question and mandatory readout
A01	R	No cue vs unrelated neutral sentence vs unrelated numeral; isolate nuisance effects from mere insertion. Pair by law, length and target.
A02	R	Numeric near-match vs far-match at fixed units and cue grammar; estimate nonlinear response and tail conditional effects.
A03	R	Exact printed-number match vs mathematical equality under alternate decimal/scientific notation; separate string identity from magnitude.
A04	R	Digit overlap matched while numerical proximity differs; test lexical-copy mechanism.
A05	R	Numerical proximity matched while digit/token overlap differs; test magnitude-comparison mechanism.
A06	R	Same-unit vs dimensionally incompatible distractor with identical numeral; test dimensional reasoning/gating.
A07	R	Same-unit vs exactly convertible-unit distractor with identical physical magnitude; distinguish text-unit from physical-unit coding.
A08	R	Explicitly mechanically isolated vs implicitly separate vs now coupled system, with the final case re-certified for relevance; test semantic causal boundary.
A09	R	Cue irrelevant to original target but relevant to counterfactual target question; test selective relevance rather than global numerical suppression.
A10	R	Cue before relevant variables, after relevant variables and adjacent to target question; test recency/position effect.
A11	R	One distractor appearance vs two and three, keeping total prompt length matched; test repetition amplification.
A12	R	Distractor as factual sentence vs quoted example vs suggested answer; test discourse role.

ID	Status	Contrast / causal question and mandatory readout
A13	C	Cue as table cell, bullet, equation, prose and code-like JSON; test representational interface without changing physics.
A14	C	Correct target zero/near-zero, very large/small, positive/negative and rounded; identify pathological relative-proximity definitions.
A15	C	Change target precision (exact irrational, 3 decimal places, 1 significant figure); measure answer-rendering ambiguity.
A16	C	Insert an unrelated intermediate quantity near a *nonfinal* intermediate answer while final answer is far; test where numerical competition occurs.
A17	C	One-step vs multi-step same final answer and comparable length; test whether internal computation depth is causal.
A18	X	Contradictory external sentence or malicious decoy; not an invariant family unless the contradiction's semantics are explicitly resolved. Quantify out-of-scope robustness separately.

6.2 B. Dataset integrity, ground truth and representation of physics (B01–B14)

ID	Status	Contrast / causal question and mandatory readout
B01	R	Free-symbol certificate vs independently programmed equation-solver certificate on a sample; disagreements go to manual adjudication.
B02	R	Symbolic mutation that introduces cue dependence; measure verifier catch rate, not just acceptance of valid equations.
B03	R	Physical-assumption mutation removing 'massless', 'frictionless', 'mechanically isolated', ideal-source or equilibrium constraints; test human and formal checks.

ID	Status	Contrast / causal question and mandatory readout
B04	R	Same target with dimensionally incorrect unit; ensure validator/parser identify error instead of silently normalizing it.
B05	R	Exact prompt hashes vs AST/graph-normalized structural duplicates; re-estimate main probe on cluster-held-out split.
B06	R	Dual expert human validation blinded to generated outputs vs single expert/agent audit; report disagreement and exclusions.
B07	R	Native answer parser vs blinded human consensus; estimate failure modes and impact on sample admission.
B08	R	Select family for scoring before vs after parseability; test selective attrition of five historically removed stable families.
B09	R	Held-out same-law new numbers vs held-out new laws/graphs; quantify actual structural generalization gap.
B10	C	Cue-type-balanced vs natural prevalence panels; quantify base-rate-driven pooled AUROC changes.
B11	C	Prompt human-naturalness blinded preference test against curated real physics exercises; reject conspicuously synthetic cues.
B12	C	Parallel physics solvers/independent human derivations for subtle thermodynamics, frame and circuit cases.
B13	C	Construct-validation error bounds: pessimistic/optimistic imputation of all unresolved family semantics; bound primary effects.
B14	X	Distinct-model content contamination probes and source provenance checks; document limits of inferring actual pretraining membership.

6.3 C. Scoring, calibration, and behavioral endpoints (C01–C18)

ID	Status	Contrast / causal question and mandatory readout
C01	R	Arbitrary base 0 vs all four rotated bases; count changed sensitivity labels without new model calls.
C02	R	Legacy signed max drop vs symmetric range/variance/MAD and no-cue-anchored signed drop; assess construct stability.
C03	R	Raw summed log-prob vs per-token average, value-only and value+unit; quantify answer-length/token effects.
C04	R	Same physical answer under multiple equivalent serialized answers; matched within-wrapper contrasts rather than raw score pooling.
C05	R	Chat template/prefix/assistant headers and exact token slicing; verify that loss only covers target output tokens.
C06	R	Gold-string likelihood vs restricted gold-vs-decoy likelihood ratio; test whether a distractor actually gains relative probability.
C07	R	Teacher-forced score vs deterministic final-answer correctness and answer-flip; expose likelihood-only outcomes.
C08	R	Multiple decoding temperatures/seeds; estimate answer distribution and reproducibility of behavioral effects.
C09	R	Fixed prompt-length matched no-cue edit vs actual cue edit; normalize against the ordinary context variation of conditional LMs.
C10	R	Next-token entropy and response-token confidence vs model-correctness calibration; no conflation between entropy and knowledge of right answer.
C11	R	Fixed FPR .05/.10, precision under shifted base rate and cost-sensitive utility; evaluate operational rather than merely ranking value.

ID	Status	Contrast / causal question and mandatory readout
C12	C	Average of valid numerical answer paraphrases vs max; quantify selection optimism and logsumexp semantic grouping approximation.
C13	C	Restricted answer-distribution JSD vs generated semantic answer divergence; specify truncation/sampling approximation.
C14	C	Confidence shift at first answer token vs along answer token chain vs first intermediate quantity; locate interface dependency.
C15	C	bf16/fp16, different batches and repeated exact prompts; show numerical accuracy of score computation.
C16	C	Different no-cue anchors matched for length/style; prove the preferred anchor is not just another arbitrary base.
C17	C	Exclude near-zero original causal sensitivities vs include with absolute-effect alternative; diagnose ratio instability.
C18	X	Alternative proper-scoring rules/semantic choice-set constructions on sufficiently enumerable answers; quantify sensitivity to answer-set selection.

6.4 D. Monitoring, statistical selectivity and model comparison (D01–D16)

ID	Status	Contrast / causal question and mandatory readout
D01	R	Four-rendering averaged L18 family probe vs strictly one-rendering deployment probe; match query budgets.
D02	R	Within-family LOOCV vs withheld 34 structural clusters and entirely unseen symbolic graph classes.
D03	R	Fitted entropy vs probe vs calibrated combined model under identical nested folds; paired difference CI.

ID	Status	Contrast / causal question and mandatory readout
D04	R	Generic correctness probe + entropy/surface vs sensitivity probe; test incremental information via held-out log loss.
D05	R	Embedding layer vs mid-layer vs final residual, preselected discovery layer vs best-of-all-layer selection.
D06	R	Digit-count, digit identity, unit lexicon, equation AST and prompt length baselines; surface models receive fair tuning.
D07	R	Class prevalence matched across cue types vs natural cue mixture; test pooled transfer inflation.
D08	R	Single prompted family vulnerability vs variant-specific event risk; avoid copying the family label into misleading variant-level claims.
D09	C	Logistic vs sparse probe vs nonlinear shallow MLP at controlled regularization and capacity.
D10	C	Hidden-state covariance/representation-stability uncertainty baselines, including multi-rendering costs.
D11	C	Calibration shift to novel laws/domains/numeral ranges and recalibration by training data only.
D12	C	Independently selected model families, reasoning-tuned checkpoints and same-backbone fine-tunes.
D13	C	Prompt-site cue-token vs last prompt vs answer-prefix states; match semantics and token positions.
D14	C	Probe directions trained on Cue A tested on B/C and vice versa; family-cluster holdout and base-rate control.
D15	C	Probe complexity/selectivity vs shuffled labels *within* structural strata and artificial metadata controls.

ID	Status	Contrast / causal question and mandatory readout
D16	X	Learning-curve and label-efficiency experiment over growing distinct template clusters, not repeated paraphrases.

6.5 E. Direct causal localization, intervention semantics and specificity (E01–E18)

ID	Status	Contrast / causal question and mandatory readout
E01	R	Qwen L16 H11-only, original four-head set, no intervention, nearby layers/heads on untouched causal families; absolute effects.
E02	R	Zero vs mean vs donor replacement vs projected subspace deletion; detect artifact of intervention distribution.
E03	R	Same layer and norm matched random direction and random head; compare dose/expected change.
E04	R	Random prompt position, relevant variable, cue numeral, cue unit, isolation token and question token; localize source-target flow.
E05	R	Matched same-answer, cue-present stable, unrelated, same-structure, different-structure donors; identify specificity.
E06	R	Correct recipient vs wrong recipient vs stable recipient; stratify by preintervention behavior and score.
E07	R	Cue near→far and far→near donor transplantation; test predicted reciprocal intervention sign.
E08	R	Knockout with/without rescue of H11 activation; test reversal of intervention effect on chosen endpoint.
E09	R	Full-head outputs vs QK pattern vs V/OV path and source-specific contribution; separate head functions.
E10	R	Accuracy/decoy-logodds vs unrelated-task loss/KL; constrain off-target damage.

ID	Status	Contrast / causal question and mandatory readout
E11	R	Edit irrelevant cue vs *relevant* numeric input that changes y*; check whether lesion destroys legitimate computation.
E12	R	Original 20 high-S families vs new unbiased and near-threshold causal panels; assess selection effect.
E13	C	Model self-repair: ablate and measure downstream activation compensation in MLP and alternate heads.
E14	C	Joint knockouts of redundant head sets vs H11 solo; estimate nonadditivity and substitute routes.
E15	C	Direction injection into stable recipients at multiple scales and sign, with off-target outcome measures.
E16	C	Cross-family activation donation aligned by physical role, source output and tokenizer position.
E17	C	Run causal ablations during teacher-forced *intermediate quantity* prediction and generation-time answer.
E18	X	Matched learned steering vectors, gradient-based erasure, linear subspace projection and feature clamping.

6.6 F. Functional mechanism, circuits, synthetic controls and mitigation (F01–F20)

ID	Status	Contrast / causal question and mandatory readout
F01	R	Head's near-match magnitude code vs same digits with distant numerical value; causal decoding and swap.
F02	R	Unit representation vs token overlap at fixed numeral; function-discriminating causal patch.
F03	R	Semantic isolation role vs no-isolation semantic variant; test whether upstream text gates H11 effect.
F04	R	Attention map vs source-token-value patch; do not infer causality from attention visuals alone.

ID	Status	Contrast / causal question and mandatory readout
F05	R	Direct logit attribution vs true nonlinear patching; compare estimated to actual intervention effects.
F06	R	Source→H11 edge patch, H11→downstream patch, knockout/rescue; validate directed causal pathway.
F07	R	Whole-residual patch vs path-specific patch, measure causal specificity and reconstruction loss.
F08	R	Retain-only-circuit vs remove-circuit; faithfulness/comprehensiveness, selected method robustness and edge-count frontier.
F09	R	Interchange intervention on high-level variable ‘cue relevant?’ vs low-level subspace; causal abstraction agreement.
F10	R	Held-out-law prediction of effect direction from discovered function; no refitting.
F11	C	Functionally aligned candidate pathway in second architecture, not same layer/head index.
F12	C	Supervised distributed alignment search vs sparse autoencoder features vs neuron/low-rank baseline; validate interchange effect.
F13	C	Synthetic planted-shortcut vs synthetic clean-trained model; estimate causal discovery sensitivity/specificity.
F14	C	Near-match-only vs multi-cue cases; test additive or competing routes.
F15	C	Mechanism-guided patch/head gate vs plain cue-removal and prompt editing; match accuracy and compute.
F16	C	Mechanism-guided training vs random distractor augmentation vs generic fine-tune on entire law-held-out test.
F17	C	Counterfactual consistency loss plus correctness loss vs either alone; test invariance/accuracy tradeoff.

ID	Status	Contrast / causal question and mandatory readout
F18	C	Adversarially generated *certified* cue attacks against frozen monitor and repair, with fresh-validation transfer.
F19	X	Small model scale/curriculum scan for emergence of numeral-copy and relevance-gating pathways.
F20	X	Open cross-layer transcoder attribution graph and feature editing; demand faithfulness under reconstruction error and alternative-feature baseline.

Total catalogue: 104 explicitly identified contrasts (18 A + 14 B + 18 C + 16 D + 18 E + 20 F), many with internal factorial cells. This is intentionally not 104 independent hypothesis tests. Group them into preregistered families, use screening/confirmation separation, and stop low-value branches early. The first-pass executable board should be roughly 12–20 **decisive experimental packages**, not 104 disconnected GPU jobs.

7 Twelve decisive experiment cards (priority order)

These cards are minimal specifications for an engineer or research agent. Each has a **primary outcome**, primary competing models, a *no-go response*, and an artifact. Do not allow the agent to invent a different estimand or panel midrun.

Card X01 — Authoritative reproduction

Inputs: committed Phase-I raw logs, benchmark manifest, model/tokenizer checksums. **Design:** independent recomputation of labels, panels, OOF probe scores, bootstrap distributions and H11 effects; validate 10+ hand-selected families directly against model forward passes. **Primary outcome:** equality to reported panel denominators, checksum and endpoint values within declared numerical precision. **Controls:** independent implementation, data/label lineage, no circular calculations from summary tables. **If failure:** block *every dependent* experiment, localize and rerun upstream. **Artifact:** signed reconciliation ledger and replayable tests.

Card X02 — Base-variant invariance challenge

Inputs: all original four-value log-prob arrays; no new GPU first. **Design:** rotate baseline through all four, compare legacy classifications to base-free range and no-cue anchor in selected examples; cross cue/model. **Outcome:** reclassification matrix and model rank concordance. **If sensitivity depends heavily on arbitrary variant 0:** retire the reference-based construct as a primary endpoint and reanalyze Phase I under preregistered replacement; retain history rather than rewriting original preregistration retroactively. **Artifact:** per-family metric multiverse with score versions.

Card X03 — Format/tokenization falsification

Inputs: stratified 60–100-family design with paired wrapper variants, both primary dense models. **Design:** 4–6 equal-meaning output wrappers and precise answer-token spans; matched cue/no-cue. **Outcome:** within-family nuisance effect versus wrapper × nuisance interaction on gold/decoy logodds and reference-free range. **If wrapper dominates:** causal-mechanism interpretation must be tied to output elicitation and cannot be universally presented as physics vulnerability. **Artifact:** tokenizer audit and hierarchical paired contrasts.

Card X04 — Physics-semantic factorial

Inputs: novel certified structures with 2 × 2 × 2 × position blocks (physical relevance × unit compatibility × digit overlap, with proximity manipulated independently where possible). **Design:** randomized cells under family/structure block, human validated, exact output protocol. **Outcome:** preregistered interaction in actual generated answer selection and gold/decoy logodds. **If no relevance-dependent effect:** prefer copying/priming mechanism and stop relevance-gating claims. **Artifact:** factorial table plus held-out conditional predicted effects.

Card X05 — Structural leakage and real OOD monitoring

Inputs: old 135 monitoring families + adjudicated 34 clusters, newly authored structure clusters. **Design:** nested train-by-cluster CV vs original family LOOCV; final frozen external test. **Outcome:** paired cluster-held-out AUROC/logloss versus fitted entropy and surface/AST baseline; CI corrected for selection. **If collapse:** monitor reads template or editing grammar; no general out-of-structure deployment claim. **Artifact:** exact fold manifest, predictions and fairness/cost table.

Card X06 — Frozen external behavioral replication

Inputs: distinct source/cluster external panel independently authored and verified, two dense models. **Design:** output-format locked, reference-free + free generation endpoints. **Outcome:** interval for cue-specific effect and model interaction, tested before mechanistic selection. **If no effect:** narrow main thesis to historic panel and investigate distribution mismatch rather than cherry-picking rare positives. **Artifact:** blind external results and null-inclusive effect distribution.

Card X07 — H11 lesion/patch specificity

Inputs: untouched Cue-B panel including near and far plus role-reversal positive controls. **Design:** H11 zero, mean, clean/corrupt donor and norm-matched controls; record all logits and free outputs. **Outcome:** absolute correct-vs-decoy logit contrast restoration with limited effect on genuinely relevant edits. **If only signed ratio improves:** causal shortcut explanation rejected. **Artifact:** per-family before/after table and layer/head stability plot.

Card X08 — Mechanistic function discrimination

Inputs: held-out factorial cells of numeric proximity, digits, units, isolation, target and placement. **Design:** QK/V/OV patching at source tokens with fixed H11 receiver; pre-register opposite sign predictions of F-copy/F-match/F-unit/F-role. **Outcome:** counterfactual effect-direction accuracy on unseen structures under genuine interventions. **If no simple function predicts:** widen to distributed mechanism; do not reverse-engineer labels from top activation examples. **Artifact:** function-by-test prediction scorecard.

Card X09 — Source-to-sink circuit and rescue

Inputs: discovery graph then separate causal test panel; independent edge discovery and true patching. **Design:** source→H11→downstream edits, knockout and rescue; faithfulness–sparsity curve with alternate ablation baselines. **Outcome:** significant selective mediated recovery and causal abstraction agreement vs random/shortcut circuit. **If source path cannot be validated:** local effect remains real at best but circuit claim removed. **Artifact:** complete edge list, paths and counterfactual intervention trace.

Card X10 — Architecture functional replication

Inputs: Llama independently selected correctly matched panels; optionally same-backbone reasoning-tuned model. **Design:** find functional analog in exploratory partition then freeze site and evaluate identical hypothesis/endpoint on independent clusters. **Outcome:** sign-consistent intervention effects for function, not same head index. **If no replication:** publish architecture-specific mechanism if deep enough; discuss alternative pathways and limited scope. **Artifact:** model-specific mechanism table and matched denominators.

Card X11 — Selective mitigation challenge

Inputs: external OOD and role-reversal panels; targeted edit, generic edit, cue removal, no intervention and solver-routing comparators. **Design:** freeze edit and gate intensity on dev; match computational costs and evaluate correct answer accuracy, correct/decoy odds, cue shift and harm. **Outcome:** constrained utility frontier and paired improvement in correct outcomes. **If simpler filter wins:** frame neural intervention as *causal diagnostic*, not deployable superiority. **Artifact:** attack/repair matrix with generated-output examples.

Card X12 — External audit and claim freeze

Inputs: untouched final cohort + full source and provenance registry. **Design:** independent review of three strongest claims, each adversarially tested with predeclared falsifier; one-time final cohort reveal. **Outcome:** precisely scoped C1–C6 claim matrix with raw reproducible artifacts and uncertainty. **If failed:** transparently contract claims or keep result as a later experiment; do not tune on external test and call it confirmation. **Artifact:** frozen manuscript evidence matrix and reproducibility bundle.

8 Statistical design: standards for inference, not decoration

8.1 Independent experimental unit, estimands and non-independence

The **independent unit** for a primary generalization claim is the independently authored *symbolic structure/template cluster*, not a rendered prompt, paraphrase, numeric instantiation, or decoder sample. A single structure may yield hundreds of observations; counting them as hundreds of independent examples is pseudoreplication. Model and cue variants are repeated measures, nested under clusters. For a generalization claim to new physics laws, treat the law/program cluster as the sampling unit or report that its population is fixed and inference is conditional on these laws. In a leave-one-domain-out test, only a few physics domains exist; do not report ordinary asymptotic p-values as if there were hundreds of independent domains.

Primary behavioral estimand (to finalize after P1 discovery): conditional average cue effect on `R_alt` (when defined) and/or correctness transition for a randomized, certified nuisance edit, averaged over a specified target distribution of physics structures and cue values. Report both population-average and cue-stratified effects. **Primary causal estimand:** on a specified panel, the average of `Y(intervention) - Y(control)` under a fixed, documented site and recipient definition; this is an intervention on *model activations*, not necessarily the natural indirect effect of cue `z` because intervening changes the computation's ordinary distribution. **Primary monitoring estimand:** out-of-sample paired difference in proper score (log loss or Brier) with AUROC as a ranking metric, conditional on the exact information/query budget.

Report both **unconditional** performance across the defined population and **conditional** performance on model-positive/sensitive cases. The second can be useful to understand a mechanism but is selected on the outcome; its result cannot establish that the effect generalizes to all physics questions. Record eligibility pre-intervention and measure no-cue/relevant control performance.

8.2 Multiple comparisons and research degrees of freedom

- Freeze a small family of confirmatory hypotheses: (H-B) behavioral factorial effect, (H-M) single-prompt incremental monitoring value, (H-C) causal head/path effect, and optionally (H-R) selective repair. Every hypothesis maps to a specific endpoint, model, held-out cohort, effect-direction prediction and null.
- Use hierarchical testing or Holm correction for the small primary family; use Benjamini–Hochberg FDR for prespecified secondary scans and annotate *exploratory* plots without pretending FDR correction makes them confirmatory.
- Selection over layers, heads, sites, formulations, decoders, metrics, threshold and cue conditions is a large multi-search. Handle it by **discovery selection on one set and evaluation once on an independent set**, not by announcing the best of many tests with a naïve CI.

- For binary effect rates, report denominator, Wilson or appropriately clustered bootstrap intervals; for model comparisons on the same families, paired contingency/paired hierarchical analyses. For continuous skewed recovery, robust median/trimmed mean, bootstrap CI, absolute numerator/denominator and sensitivity to few extreme families are mandatory.
- The classifier's calibration binning is itself estimated; show reliability curves and uncertainty with sufficient bin support, and report ECE with fixed, declared bins plus a supplementary adaptive strategy. Never infer 'best calibrated' solely from Brier or solely from ECE.

8.3 Power and precision protocol

Before collecting confirmation labels, simulate from pilot-estimated family-cluster variance and cue-specific distributions. Sweep realistic numbers of independent structures, 2–8 cue conditions, 2–3 numeric instantiations and models. Compute estimated CI width, type-I error with the chosen resampling unit and power for minimum scientifically important contrasts. Run the *exact proposed analysis* on simulated null and effect datasets; ensure false-positive rate is calibrated. Include a scenario with 20–40% loss from validation, parse failures, and ineligible causal recipients. An analysis that is too underpowered for ±0.03 AUROC improvement must not make a 'statistically equivalent' claim; use an equivalence test only after specifying a minimum practically important margin and achieving sufficient power.

Sample size is decision-dependent. Starting target of 300–600 new symbolic structures is a planning allocation for independent validity and power, **not** a claim that this number will resolve all comparisons. A tiny difference in paired AUROC may require much more than 600 if cluster correlation and label prevalence are unfavorable. Conversely, a large, specific causal effect could require fewer untouched structures if all other controls are convincing. Publish the simulation and its assumptions alongside the final chosen number.

8.4 Negative and missing result policy

A zero-row job, failed hook, timeout, nonparseable response, ineligible family, or untested control is `NOT_EVALUATED`, not a null effect. Keep a missingness codebook with `reason`, `first_seen`, `planned_run`, `attempts`, `eligible_population`, and intended analysis. Report exclusions and completeness both by cue and by model. If a test fails after looking at response scores, keep all original units in a sensitivity analysis, including worst-case bounds for missing outcomes. Never silently replace data-dependent missing values with zero or negative labels.

9 Compute, human effort, storage and reproducibility architecture

Hardware assumption to verify. The June proposal described access patterns of up to **2×H200, 3×H100, 2×A100**, not a guaranteed simultaneous dedicated allocation. Before beginning, check currently available scheduler partitions, actual per-job GPU limits, software stacks and wall-time quota. Specific throughput and costs must be measured on Sharanga rather than invented here. Models should be reselected against present availability, licensing and stable hook support; avoid committing to a 397B-class model merely because an old proposal mentioned one.

Cost model. For behavioral runs, estimate `N_structures × N_numeric × N_cue_cells × N_models × (N_score_forward + N_generated_resamples)` and record separately the repeated output-format wrappers. For activation jobs, additionally multiply by selected layer/site/head/position sets and donor-recipient pairs; cache only what is actually reused. One full all-layer/all-token dump can grow to terabytes for large models; the submitted manuscript cites a 1.57 TB upper-bound design motivating staged extraction. Recompute memory from `n_items × n_positions × n_layers × d_model × bytes_per_value` for current models rather than copying this figure.

Compute escalation:

1. **CPU and light GPU:** P0 registry, exact-raw logprob rescoring where already stored, parser and symbolic mutation, cluster graph; no model change until integrity passes.

2. **Small balanced GPU pilot:** P1 wrapper and C contrast, 30–100 families, two models; estimate rates and precision and debug tokenizer/hook parity.
3. **Benchmark creation/validation:** P2/P3 generation with symbolic solver and human review, parallel to small discovery not to locked test scoring.
4. **Behavioral confirmatory sweep:** only validated families and frozen output protocol; favor inexpensive inference batching.
5. **Targeted activation discovery:** 20–60 *independent* contrast pairs per functional hypothesis rather than storing all activations on every prompt.
6. **Causal confirmation:** freeze 3–10 meaningful candidate sites/pathways from discovery, then run sufficiently powered held-out interventions with stringent controls.
7. **Optional heavy:** multiple architecture causal studies, trained SAE/transcoder, intervention-based fine-tuning, adversarial optimization, if mechanism survives earlier gates.

Storage guidance. Persist normalized per-family/per-variant *raw records* and complete model provenance; store activation caches transiently in scratch with object index/hashes; preserve only discovery-critical activation slices and reproducible extraction scripts if space constrained. Use `safetensors`/zarr/HDF5 or a tested equivalent, avoid pickled arbitrary code in shared artifacts, pin dtype, shape and named hook coordinates. Log when a source cache was produced under an earlier model revision. Make every run restartable and idempotent.

Human validation workload. Conditional on an approved Phase-B scaling decision, for 400 distinct structures with two validators, budget at least 800 independent family-level judgments *plus* calibration/adjudication, and more if every numeric/linguistic variant needs separate review. Record average adjudication time during a 20-family pilot, then compute workload empirically. Determine qualified validator criteria, impartial assignment, pay, conflicts, communication and completion milestones. A self-authored agent review is not independent expert validation.

First-stage staffing. One investigator owns scientific hypotheses, effect definitions and claim ledger; one owns benchmark/symbolic verifier; one owns model instrumentation and reproducibility; one coordinates validators; at least one independent reviewer attacks data leakage and causal interpretation. Roles can be combined if headcount is small, but independence checks cannot all be performed by the same agent that wrote the original data.

9.1 Machine-readable data contracts

Every raw record has `run_id`, `phase_id`, `exploratory_or_confirmatory`, `structure_uuid`, `template_cluster_id`, `symbolic_program_hash`, `cue_type`, `cue_semantic_role`, `cue_numeric_value`, `unit_type`, `numeric_proximity`, `digit_overlap`, `cue_position`, `isolation_condition`, `model_id`, `model_commit`, `tokenizer_commit`, `chat_template`, `answer_wrapper_id`, `target_answer_ids`, `target_span_start/end`, `generated_answer_raw`, `parsed_answer`, `parse_status`, `gold_logprob_raw/per_token`, `alternative_scores`, `correctness`, `seed`, `dtype`, `wall_time`, and `artifact_checksum`. For causal results add `receiver_id`, `donor_id`, `edit_site`, `edit_semantics`, `scale`, `clean/corrupt_definition`, `hook_tensor_shape`, `absolute_Y_control`, `absolute_Y_intervened`, `KL_damage`, `model_positive_eligibility`, and recipient generation outcomes. Every derived table names the exact source manifest and calculation commit.

Continuous integration checks: invariant answer across nuisance edits; correct answer changes under certified relevant positive controls; all cue spans are the only allowed mutations in restricted cells; exact duplicate and cluster split integrity; correct target span IDs; no train/test family or cluster collisions; leakage-free preprocessing; out-of-fold score generation; identical x-axis/denominators for paired models; no zero-row falsely interpreted as zero effect; hooks removed after each run; no gradients accumulated inadvertently; no confirmed run using a development-fitted threshold on external labels.

9.2 Preflight notebook skeleton / pseudocode

```
# Conceptual validation contract, not runnable Phase-I implementation.
manifest = freeze_structural_splits(certified_and_human_checked_families)
assert no_cluster_overlap(manifest.discovery, manifest.confirmation)
```

```
assert no_cluster_overlap(manifest.confirmation, manifest.external)

for model in selected_models:
    validate_native_chat_template(model)
    validate_scored_answer_span(model)
    validate_hook_parity(model)
    for group in manifest.discovery:
        records = score_and_generate_all_preregistered_cells(model, group)
        assert cue_preserving_answer_invariance(records.symbolic_templates)
        emit_immutable_raw_records(records)

# Freeze choices using discovery ONLY.
protocol = choose_primary_metric_model_site_hypothesis(manifest.discovery)
write_preregistration(protocol, git_sha=current_sha())

# Confirmation panel is loaded ONCE by frozen script.
result = evaluate_frozen_protocol(protocol, manifest.confirmation)
report_cluster_robust_effects_and_all_eligible_units(result)

# Unsuccessful results never silently reroute to a new confirmatory test.
if not result.passes_prespecified_gate:
    close_claim_and_open_new_discovery_branch(protocol, result)
```

10 Conditional decisions: how to react if the evidence is unfavorable

Gate and observed outcome	Scientific interpretation	Required action	Prohibited reaction
P0 raw outputs not reproducible	Evidence chain broken or model/runtime drift	Fix code/provenance, rerun upstream and invalidate dependents	Continue adding heads/plots around unverified metrics
P1 scores change with base rotation	Current binary label not indexing-invariant	Use anchor-free and explicitly anchored endpoints; reassess paper's claims	Relabel retrospectively and say the original preregistration always meant the new metric
P1 wrapper/tokenization explains effect	String-scoring artifact or strong interaction	Restrict claim; study interface sensitivity as Track E	Assert physics reasoning failure from gold logprob alone
P1 generated answers stable, p(y*) drifts	Confidence/likelihood-only phenomenon	Focus reliability/calibration, not answer corruption	Claim model accuracy is repaired by likelihood edit
P2/3 human review rejects semantics	Benchmark construct invalid for affected prompts	Redesign and revalidate, rerun affected findings	Treat solver free-symbol check as sufficient evidence of natural-language correctness
P4 no relevance × proximity interaction	Physical relevance gating unsupported	Test numeral-copy/generic-context alternatives	Add post-hoc cues until desired sign appears
P4 strong Cue B, weak Cue A/C	Heterogeneous cue phenomenon	Deep-dive B; A/C as scope tests	Pool cue types to claim a universal mechanism
P5 monitor fails cluster split	Template recognizer more plausible than universal monitor	Remove deployment superiority; focus function or new representation only after fresh discovery	Quote best family-LOOCV AUROC as OOD score

Gate and observed outcome	Scientific interpretation	Required action	Prohibited reaction
P5 matches entropy under adequately powered test	Hidden states add little incremental value for this task	Report negative and avoid costly deployment claim	Interpret absence of significance as superiority or equivalence without proper test
P6 H11 zero only, no mean/patch effect	Off-manifold lesion or delicate compensation	Diagnose lesion; withhold causal function claim	Call H11 necessary in all circumstances
P6 H11 effect is gold likelihood only	Causal contribution to scoring distribution	Pursue specific output-sensitivity account or other metrics	Claim improved physics solution computation
P7 no one-feature account	Distributed or multiple mechanism possible	Test subspaces, multiple heads, edge discovery; simplify scope	Human-label top activating examples and declare mechanistic understanding
P8 path attribution fails rescue	Proposed information route wrong or insufficient	Redefine pathway on discovery data or fall back to localization	Publish attribution arrows as proven information flow
P9 second model has different component	Functional mechanism may still recur	Find analog with independent discovery/confirm, or narrow claim	Equate same-layer misses with true architectural absence
P9 no external functional replication	Architecture/benchmark-specific mechanism	Limit generalization, strengthen Qwen case study and explain boundary	Keep universal LLM language
P10 cheap cue filter dominates	Circuit repair not practical	Treat edit as scientific intervention, not product method	Exclude cheap filter from comparison
Final external set contradicts preferred claim	Proposed mechanism/family not universal	Report boundary, stop tuning, collect new confirmation set for revised claim	Reuse opened external cohort until passing

Stop–pivot hierarchy: credibility of the measured effect outranks benchmark size; causal interpretation outranks component count; reproducible out-of-cluster prediction outranks peak AUROC; selective correctness improvement outranks ratio recovery. The plan is adaptive about which *branch* gets investigated, not about rewriting confirmatory statistics after outcomes.

11 Literature-grounded positioning and novelty test

The following is a **source-to-design map**, not a list of papers inserted to make the proposal appear scholarly. Links were located and cross-checked against publisher, author-project or arXiv sources as of **14 October 2026**. Publications marked `preprint` were not independently verified as peer-reviewed here. Every referenced paper must be read in full by the investigators before using it to claim novelty; this document gives the concrete method/overlap justified by the source's verified abstract/method description. In particular, **absence of a related work in this selective review is not proof that no similar paper exists**.

Literature	What is established or proposed	Implication for PhysMon and decisive differentiation
[L01] Mirzadeh et al., *GSM-Symbolic* (ICLR 2025)	Symbolic templates vary arithmetic problem instantiations and demonstrate failure under perturbations/irrelevant clauses.	Dataset novelty is **not** 'symbolic problem generation'. Our incremental claim must concern certified *physics-specific* nuisance invariances, multiple behavioral estimands and an experimentally validated internal pathway.
[L02] Yang et al., *GSM-DC* (EMNLP 2025)	Synthetic symbolic reasoning graphs with controlled irrelevant context, evidence about paths/accuracy, and strong-distractor robustness training.	Very close benchmark and mitigation competitor. Compare controlled graph robustness, distractor strength, and augmentation; do not say prior work lacks controlled symbolic distractors.
[L03] Cheng et al., *Stochastic Chameleons* (ACL 2025)	Internals of irrelevant-context hallucinations and competing pathways across Llama/Mistral/Pythia in factual recall.	Two-path mechanism is **not** unprecedented. Need a novel physics-grounded causal variable/function, not just generic 'two circuits compete'.
[L04] Nikankin et al., *Arithmetic Without Algorithms* (ICLR 2025)	Sparse arithmetic heuristic neurons implement numerical pattern-specific behaviors.	Numerical near-match effects may be an arithmetic heuristic; compare numeric pattern/function tests before invoking scientific reasoning.
[L05] Zhang & Nanda, *Best Practices of Activation Patching* (ICLR 2024)	Patch outcomes depend on corruption, metric and intervention choices.	Justifies absolute logit/likelihood endpoints, alternative ablations, clean/corrupt protocols, and non-inflated normalized recovery.
[L06] Miller et al., *Circuit Faithfulness Metrics Are Not Robust* (preprint 2024)	Reported circuit faithfulness is sensitive to ablation definitions.	Requires complement restoration, multiple ablation semantics and explicit boundary of the circuit claim.
[L07] Geiger et al., *Causal Abstraction* (JMLR 2025)	Formal framework connecting low-level intervention, abstract variables, interchange and graded faithfulness.	Provides formal target for *what a purported relevance-gating subspace computes*.
[L08] Mueller et al., *MIB* (ICML 2025)	Benchmark comparing circuit and causal-variable localization; supervised DAS strong, SAE not uniformly superior to neurons.	Evaluate discovered pathways on causal-variable/faithfulness tests; do not assume SAEs automatically deepen interpretation.
[L09] Syed et al., *Attribution Patching Outperforms Automated Circuit Discovery* (preprint 2023)	Gradient attribution can efficiently shortlist important edges.	Use as inexpensive candidate discovery, followed by full intervened evaluations.
[L10] Bricken et al., *Circuit Tracing* (2025 technical paper)	Feature-level attribution graphs; limitations include reconstruction error and attention/global-circuit ambiguity.	Attribution graph is a hypothesis generator; show causal path/knockout/rescue and reconstruction controls.
[L11] Lindsey et al., *On the Biology of a Large Language Model* (2025 technical paper)	Detailed case studies using circuit-tracing methodology.	Benchmark qualitative depth of functional analysis, not claimed as a direct comparison on PhysMon.
[L12] McGrath et al., *Hydra Effect* (preprint 2023)	Ablating attention components can cause compensating computations downstream.	A zero-ablation result may not identify normal unperturbed information flow; measure self-repair.

Literature	What is established or proposed	Implication for PhysMon and decisive differentiation
[L13] Kossen et al., *Semantic Entropy Probes* (preprint 2024)	Predict sampling-derived semantic uncertainty cheaply from hidden states.	Stronger uncertainty baseline where task labels and compute budgets make evaluation meaningful.
[L14] Chen et al., *INSIDE* (ICLR 2024)	Hidden-state covariance/eigenvalue statistics for hallucination detection.	Adapt comparable representational uncertainty controls without conflating multiple-generation access with one pass.
[L15] Khanmohammadi et al., *CCPS* (EMNLP 2025)	Perturbed representation stability improves confidence calibration on QA benchmarks.	A serious internal-state competitor; its actual cost and perturbation protocol must be matched rather than treated as one free forward pass.
[L16] Sun et al., *Emperor's New Clothes in Benchmarking?* (ICML 2025)	Controlled assessment of fidelity vs contamination resistance across benchmark modifications.	Merely paraphrasing old DECOY templates may not yield independent new tasks; demand structural novelty.
[L17] Simin Chen et al., *Benchmarking Under Data Contamination* (EMNLP 2025)	Survey of static/dynamic benchmark contamination risks.	Include independent source/cluster tests and external benchmark provenance.
[L18] Chen, Luo & Pan, *Mechanistic Data Attribution* (ICML 2026)	Tests training-data contributions to formation of interpretable circuits.	Training-origin tracing is an optional ambitious **different** question; avoid claiming mechanism originated from particular pretraining examples without actual provenance/intervention.
[L19] Khan et al., *Can LM Agents Be Helpful Circuit Explainers?* (preprint 2026)	Agentic explanation workflow can be useful but validation failures remain a challenge.	A coding agent may propose hypotheses, but experimental falsification and human ownership are mandatory.
[L20] Naganna et al., *Are Arithmetic Heuristic Neurons Form-Invariant?* (preprint 2026)	Studies arithmetic-neuron transfer across symbols, word problems and code.	Strong motivation for format-invariance tests of numeric components; independently verify final publication status before citing as peer reviewed.
[L21] Geirhos et al., *Shortcut Learning in Deep Neural Networks* (Nature MI 2020)	General conceptual foundation for shortcuts and distribution shift.	Use precise shortcut language; no novelty for demonstrating that shortcuts exist.
[L22] Heimersheim & Nanda, *How to Use and Interpret Activation Patching* (preprint 2024)	Practical pitfalls of interpreting patched activation effects.	Required reading before designing causal sites/donors; not itself a proof of any proposed pathway.
[L23] OpenAI/TransformerLens code (`patching.py`)	Provides generic activation and head/position patching interfaces.	Implementation reference, not a scientific claim; validate actual hook tensor identities and attention-head indexing.
[L24] Anthropic open-source circuit-tracing announcement (2025)	Public tooling for graph-based circuit exploration on open weights.	Optional artifact/visualization tool, not substitute for P8 causal confirmation.
[L25] ICLR 2027 Reviewer Guidelines	Emphasize motivation, technical soundness, supported claims and significance; SOTA leaderboard dominance is not a requirement.	Target new knowledge and explanatory rigor rather than experiment count.

11.1 What could actually be novel (conditional claims)

N1: A validated, reference-independent measurement of physics-specific nuisance influence that jointly captures gold-answer probability, distractor-answer competition and generated-answer behavior, with explicit separation of language-model scoring artifacts. **Risk:** GSM-DC and GSM-Symbolic already offer strong controlled benchmarks; a new metric alone may not suffice.

N2: A *mechanistically predictive* function for irrelevant numerical information (e.g., a quantified relevance-gating or numeral-copy pathway) established by orthogonal cue factorials, knockout, reciprocal transplantation, path patching, rescue and causal abstraction on unseen physics computation graphs. **Risk:** broader class-based misgeneralization and arithmetic heuristics are already known; the specific functional discovery must exceed them.

N3: A rigorous demonstration that a single-prompt internal monitor adds (or does **not** add) useful information beyond properly nested fitted uncertainty, correctness and numerical-template baselines, with independent cluster/source transfer. **Risk:** a small unresolvable AUC difference, by itself, is not a major contribution.

N4: A selective intervention that changes the identified mechanism's contribution and improves *correctness/reliability* under valid nuisance edits without disabling legitimate use of relevant variables, outcompeting inexpensive alternatives on some defensible frontier. **Risk:** a raw accuracy boost from fine-tuning without mechanistic linkage belongs to robust training, not necessarily mech interp.

Novelty falsification task before submission. Re-run targeted literature searches for `certified physics distractor`, `counterfactual shortcut circuit`, `numerical competition attention heads`, `causal relevance gating`, `causal abstraction scientific reasoning`, `arithmetic heuristic mechanism` and `confidence variation answer string`; inspect papers' experiment tables and code. Maintain a `closest_work_matrix.csv` with year/venue, exact benchmark generator, formal guarantee, behavioral target, models, localization method, functional proof, external validity and intervention. No phrases like 'first to show' unless directly verified against the full adjacent literature.

12 The first execution sprint: benchmark credibility before expensive mechanism work

These actions are ordered by which uncertainty could make all later experiments meaningless. Do not treat the order as calendar deadlines. A coding agent may execute independent audit jobs in parallel, but later **confirmatory** jobs remain gated.

1. **Tag and reproduce Phase I.** Freeze the *Confidence Without Cause* 25-page ICLR PDF, 155-family manifest, model/tokenizer/chat-template commits, cue-parent lineage and every raw model record. Distinguish 140 original behavioral families from 15 later expansion families and the 135-family monitor cohort. **Exit:** a top-to-bottom regenerated Phase-I claim table and list of conflicts.
2. **Run the zero-cost baseline permutation attack.** For every four-rendering family, rotate the base index through all possibilities; compute the signed legacy score and a base-free range. Tabulate label reclassification, Cue-A/B/C stratification and cross-model ranking. **Exit:** self-contained unit tests, machine-readable `base_permutation_audit.csv` and a reviewable discrepancy ledger.
3. **Inspect the scoring interface.** Verify prompt bytes, answer-token spans, model-native templates and manual teacher-forced scoring on representative near-match, unit-conversion, negative-score, wrong-answer and parse-failure cases. Compare reference answer formats and numerical rounding. **Exit:** signed `scoring_contract.md` with passing/failing cases.
4. **Resolve structure overlap and the parser cohort issue.** Independently adjudicate the 34 reported structural clusters and 2,352 unresolved rendering pairs; rerun cluster-held-out baselines. Identify why five stable families were excluded from teacher-forced monitoring. **Exit:** frozen lineage graph and eligibility trace with nonselected families retained.
5. **Qualify independent physics validators.** Build a two-pass blind rubric and planted-invalid calibration bank. Audit a stratified sample of v1 large/small effects and Cue A/B/C, then budget

the full audit. **Exit:** qualification decisions, agreement, defect taxonomy, reviewer workload and adjudication protocol.

6. **Implement the typed relevant/irrelevant generator.** Create 10 fully certified new physical structures, each with no-cue, relevant-edit, non-governing, numeric metadata and distractor magnitude/unit conditions; automatically generate mutation tests. **Exit:** reproducible generator, admissibility graph, factor-cell manifest and genuinely independent expert review.
7. **Pilot approximately 40–60 independent structures.** On two open models, test the core relevance × proximity × unit interaction with free-answer behavior, right-versus-decoy log-odds and symmetric gold-string dispersion. Test near-match vs token-overlap confounding. **Exit:** discovery-only effect sizes, uncertainty, failures and a clear preferred or rejected behavioral hypothesis.
8. **Pilot external adaptation independently of model outputs.** Draw prespecified candidate pools from PHYSICS-Verified and UGPhysics, reconstruct the physics, certify nuisance/relevant edits and record all exclusions. Add a smaller PhysReason feasibility check if resources allow. **Exit:** empirical adaptation yields, validation hours, task difficulty and a decision on EPC feasibility.
9. **Power and freeze the benchmark.** Use pilot-derived cluster variance and eligible/attrition rates to set the v2 size and confirmation split. Retire Cue C from irrelevant-cue pooling, document the separate invariance/equivariance benchmark, set confirmed endpoints and preregister tests. **Exit:** immutable Phase-B gate report `GO—MECHANISM / GO—MEASUREMENT / PIVOT—GENERAL NUMERIC / REDESIGN / STOP BROAD CLAIM`.
10. **Only on `GO—MECHANISM`, start large-scale behavior and mechanistic confirmation.** Run fresh frozen model comparisons on approved internal/external panels. Re-test Qwen H11 using absolute logits/answer competition, multiple ablation semantics and matched relevant edits before planning upstream path tracing. **Exit:** results that directly justify or prohibit the later circuit hypothesis.
11. **Write claim-specific experiment cards and update resource estimates.** Freeze exact eligibility rules, models, hook sites, donors, statistics, cluster bootstrap plan, human-review status, GPU budget and failure branches. **Exit:** reproducible queue and independently reviewable claims-to-evidence matrix.

The key policy is to allow a credible early negative result to change the project, rather than allowing the initial head-11 story to determine which benchmark we build.

13 Proposed paper architectures depending on the evidence

Architecture A: *A Causal Mechanism of Numerically Induced Physics Reasoning Instability* (high bar)

Core figure chain: certifiable relevance factorial → behavioral source of vulnerability → causal head/function dissection → upstream path and knockout/rescue → cross-architecture analog → selective control. **Required:** truly new mechanistic content, causal abstraction or similarly strong functional tests, and external structural confirmation. **Cannot be claimed from:** one head knockout, an attractive attention map, high probe AUROC, or a correlated direction.

Architecture B: *What Do Certified Counterfactuals Actually Measure?* (high-quality measurement route)

Core chain: original label ambiguity → format/base/tokenization audit → correct/decoy/free-generation alternative endpoints → new proof-carrying factorial benchmark → stringent cluster splits and model comparisons → limits of one-input monitoring. **Required:** surprising and consequential empirical evidence plus a generalizable diagnostic protocol. **Cannot be claimed from:** only more DECOY problem variants or a minor change in AUROC.

Architecture C: *Mechanism-Guided Selective Robustness to Physically Irrelevant Information* (intervention route)

Core chain: causal vulnerability → verified specific mechanism → cheap and circuit-aware repair baselines → held-out relevant/irrelevant role reversals → quality/compute/damage frontier.

Required: reliable performance gains on genuine correctness/reliability outcomes, external generalization, meaningful tradeoff improvement and causal attribution. **Cannot be claimed from:** merely flipping the teacher-forced score or an untested rule-based cue filter.

Editorial discipline. A future main paper should have one decisive thesis with appendices supporting it. A research programme may discover three publishable projects; forcing all into one giant paper often weakens explanatory clarity. A future conference could reject excellent work for novelty/clarity reasons; this document promises a scientifically credible process, not an acceptance probability.

14 Bibliography and primary-source links (verified online 14 Oct 2026)

1. **[L01]** I. Mirzadeh et al. (2025). *GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models*. ICLR. https://proceedings.iclr.cc/paper_files/paper/2025/hash/ec2e7a896f8250986b3907f57621ce94-Abstract-Conference.html
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3. **[L03]** Z. Cheng et al. (2025). *Stochastic Chameleons: Irrelevant Context Hallucinations Reveal Class-Based (Mis)Generalization in LLMs*. ACL, pp. 30187–30214. https://aclanthology.org/2025.acl-long.1458/
4. **[L04]** Y. Nikankin et al. (2025). *Arithmetic Without Algorithms: Language Models Solve Math with a Bag of Heuristics*. ICLR. https://proceedings.iclr.cc/paper_files/paper/2025/hash/8c5f30296296d2ae402ebbd09aaa9c12-Abstract-Conference.html
5. **[L05]** F. Zhang and N. Nanda (2024). *Towards Best Practices of Activation Patching in Language Models: Metrics and Methods*. ICLR. https://proceedings.iclr.cc/paper_files/paper/2024/hash/06a52a54c8ee03cd86771136bc91eb1f-Abstract-Conference.html
6. **[L06]** J. Miller, B. Chughtai, W. Saunders (2024). *Transformer Circuit Faithfulness Metrics are not Robust*. arXiv preprint. https://arxiv.org/abs/2407.08734
7. **[L07]** A. Geiger et al. (2025). *Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability*. JMLR 26(83):1–64. https://www.jmlr.org/papers/v26/23-0058.html
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10. **[L10]** T. Bricken et al. (2025). *Circuit Tracing: Revealing Computational Graphs in Language Models*. Transformer Circuits technical report. https://www.transformer-circuits.pub/2025/attribution-graphs/methods.html
11. **[L11]** J. Lindsey et al. (2025). *On the Biology of a Large Language Model*. Transformer Circuits technical report. https://www.transformer-circuits.pub/2025/attribution-graphs/biology.html
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13. **[L13]** J. Kossen et al. (2024). *Semantic Entropy Probes: Robust and Cheap Hallucination Detection in LLMs*. arXiv preprint. https://arxiv.org/abs/2406.15927
14. **[L14]** C. Chen et al. (2024). *INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection*. ICLR. https://mlanthology.org/iclr/2024/chen2024iclr-inside/
15. **[L15]** R. Khanmohammadi et al. (2025). *Calibrating LLM Confidence by Probing Perturbed Representation Stability*. EMNLP. https://aclanthology.org/2025.emnlp-main.530/

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23. **[L23]** TransformerLens (repository implementation). *patching.py*. https://github.com/TransformerLensOrg/TransformerLens/blob/main/transformer_lens/patching.py

24. **[L24]** Anthropic (2025). *Open-sourcing Circuit Tracing Tools*. https://www.anthropic.com/research/open-source-circuit-tracing

25. **[L25]** ICLR 2027. *Reviewer Guidelines* and *Author Guidelines*. https://iclr.cc/Conferences/2027/ReviewerGuidelines ; https://iclr.cc/Conferences/2027/AuthorGuidelines

26. **[L26]** Yale NLP. *PHYSICS-Verified: Audited Advanced Physics Benchmark* (public dataset card; inspected 14 October 2026). 1,034 qualifying-exam problems, 2,559 scored answers; newly adapted counterfactuals are **not** automatically validated by this source. https://huggingface.co/datasets/yale-nlp/physics-verified

27. **[L27]** X. Xu et al. (2025). *UGPhysics: A Comprehensive Benchmark for Undergraduate Physics Reasoning with Large Language Models*. ICML, PMLR 267:69849–69877. https://proceedings.mlr.press/v267/xu25ai.html ; official repository: https://github.com/YangLabHKUST/UGPhysics

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30. **[L30]** Yale NLP. *PHYSICS Benchmark Repository and PHYSICS-Verified migration guidance* (audited source/reuse notes; inspected 14 October 2026). https://github.com/yale-nlp/Physics

Attached-source index (not public citations): Actual submitted manuscript: *Confidence Without Cause* (25 pp., October user-provided); June original proposal (30 pp.); supplementary change log (14 pp.); historical June Part-II plan v1.1 (42 pp.); supplied Google PAT/OpenReview LLM review (5 pp.); separate 29 pp. *Proved, Not Annotated* manuscript of undetermined relationship to submission. These primary artifacts motivate the initial evidence ledger and critiques; **this document is a newly proposed protocol, not an attestation that suggested studies have been run**.

15 Definition of scientific completion

A final ICLR-calibre submission is scientifically ready **only after Phase B passes its applicable gate**, and only when it can answer all of the following, with independently verifiable artifacts: **(1)** What precise model behavior changed, under which definition, and why should the physical answer be invariant? **Which separate tests independently establish natural-language physical validity and matching of the encoded graph?** **(2)** What weaker scoring/format/difficulty hypothesis was strongest, and where was it falsified? **(3)** Did a monitor add genuine information beyond fair inexpensive baselines on unseen structures, and if not is this acknowledged? **(4)** What neural computation was identified beyond an effect of removing a head? **(5)** Which controlled intervention predictions would fail if the proposed mechanistic story were false, and what happened when tested? **(6)** Which models, cue types, structures and prompting conditions do and do not replicate? **(7)** Can intervention improve a meaningful outcome without disrupting legitimately relevant numerical information? **(8)** Can an independent researcher reproduce both successes and null results, including denominators and exclusions? **(9)** Is the core insight substantially different from GSM-DC, GSM-Symbolic, numerical-heuristic circuits and existing causal interpretability work? **(10)** Does the manuscript say no more than the evidence supports?

If the answer to (4) or (5) is no, the project can still produce rigorous measurement science, but it should not market itself as a completed mechanistic explanation. If (1) fails, return to the measurement definition regardless of how impressive other metrics look.